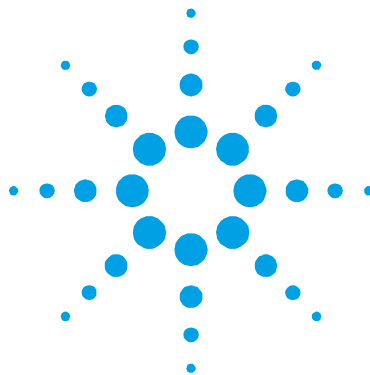




TwisTorr 74 FS

Pump models: X3502-64170, X3502-64171,
X3502-64172, X3502-64173

Manuale di istruzioni
Bedienungshandbuch
Notice de mode d'emploi
Manual de instrucciones
Manual de instruções
Bedrijfshandleiding
Istrukstionsbog
Bruksanvisning

Instruksjon manual
Ohjekäsikirja
Felhasználói kézikönyv
Podrecznik instrukcji
Návod k použití
Návod na obsluhu
Priročnik za navodila
User Manual

87-901-053-01
07/2018



Agilent Technologies

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CAUTION

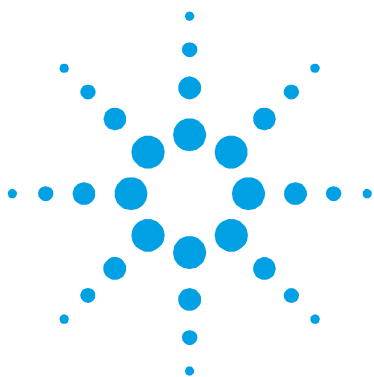
A **CAUTION** notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a **CAUTION** notice until the indicated conditions are fully understood and met.

WARNING

A **WARNING** notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in personal injury or death. Do not proceed beyond a **WARNING** notice until the indicated conditions are fully understood and met.

TwisTorr 74 FS Pump





16 Instructions for Use

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Original instructions



Safety Guideline for Turbomolecular Pumps

Turbomolecular pumps as described in the following operating manual contain a large amount of kinetic energy due to the high rotational speed in combination with the specific mass of their rotors.

In case of a malfunction of the system for example rotor/stator contact or even a rotor crash the rotational energy may be released.

WARNING!



To avoid damage to equipment and to prevent injuries to operating personnel the installation instructions as given in this manual should be strictly followed!

General Information

This equipment is destined for use by professionals. The user should read this instruction manual and any other additional information supplied by Agilent before operating the equipment. Agilent will not be held responsible for any events occurring due to non-compliance, even partial, with these instructions, improper use by untrained persons, non-authorized interference with the equipment or any action contrary to that provided for by specific national standards.

The TwisTorr 74 FS is a turbo-molecular pump for high and ultra-high vacuum applications, which is able to pump any type of non corrosive gas or gas compound. It is not suitable for pumping liquids or solid particles.

The pumping action is obtained through a high speed turbine (max. 70000 rpm) driven by a high performance 3-phase electric motor. The TwisTorr 74 FS is free of contaminating agents and, therefore, is suitable for applications requiring a "clean" vacuum.

The following paragraphs contain all the information necessary to guarantee the safety of the operator when using the equipment. Detailed information is supplied in the appendix "Technical Information".

This manual uses the following standard protocol:

WARNING!



The warning messages are for attracting the attention of the operator to a particular procedure or practice which, if not followed correctly, could lead to serious injury.

CAUTION!

The caution messages are displayed before procedures which, if not followed, could cause damage to the equipment.

NOTE

The notes contain important information taken from the text.

Safety Symbols

SYMBOL	SYMBOL DESCRIPTION
	The apparatus is marked with this symbol when the user should refer to the instruction manual in order to protect risk of harm to the operator and to protect the apparatus against damage.
	The apparatus is marked with this symbol when the user should stay clear during operation of the apparatus.
	The apparatus is marked with this symbol when the user has to connect a protective earth connection to the apparatus.

Storage

In order to guarantee the maximum level of performance and reliability of Agilent Turbomolecular pumps, the following guidelines must be followed:

- when shipping, moving and storing pumps, the following environmental specifications should not be exceeded:
 - temperature range: -40 °C to 70 °C
 - relative humidity range: 0 to 90 % (non condensing)
- the turbomolecular pumps must be always soft-started when received and operated for the first time by the customer
- the shelf life of a turbomolecular pump is 12 months from the shipping date.

CAUTION!

If for any reason the shelf life time is exceeded, the pump has to be returned to the factory. Please contact the local Agilent Vacuum Sales and Service representative for informations.

Preparation for installation

The TwisTorr 74 FS pump is supplied in a special protective packing. If this shows signs of damage which may have occurred during transport, contact your local sales office.

When unpacking the pump, be sure not to drop it and avoid any kind of sudden impact or shock vibration to the TwisTorr 74 FS pump.

Do not dispose of the packing materials in an unauthorized manner. The material is 100% recyclable and complies with EEC Directive 94/62 and subsequent amendments.

CAUTION!

In order to prevent outgassing problems, do not use bare hands to handle components which will be exposed to vacuum. Always use gloves or other appropriate protection.

NOTE

Normal exposure to the environment cannot damage the TwisTorr 74 FS . Nevertheless, it is advisable to keep it closed until it is installed in the system, thus preventing any form of pollution by dust.

16 Instructions for Use

Preparation for installation

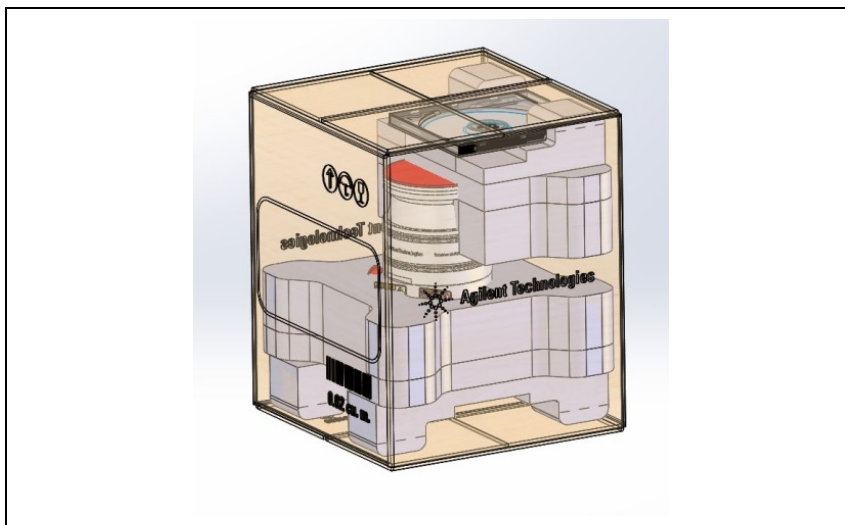


Figure 1 X3502-64170, X3502-64171, X3502-64172, X3502-64173

Installation

WARNING!



The pump is running at high speed. Please install the pump first before running it. Operating the pump without connecting it to your system gives access to the rotating parts which may cause heavy injuries.

WARNING!



The power cord of the controller is defined as disconnecting device in order to remove the power from the pump and the controller in case of an emergency. Make sure the power connector of the instrument can be easily reached and unplugged. Provide sufficient space behind the power socket of the controller to unplug the cable.

If the controller cannot be installed in the system, so that the power cord can be easily reached, a different option has to be installed to disconnect the controller from mains.

CAUTION!

Do not remove the adhesive and protective cap before connecting the turbopump to the system.

16 Instructions for Use Installation

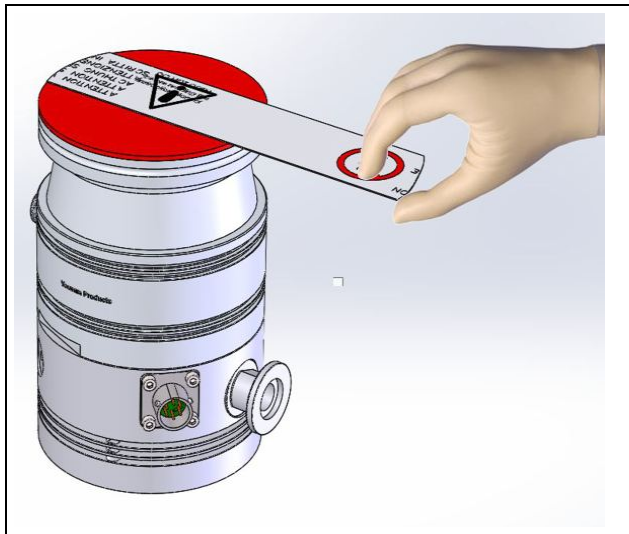


Figure 2



Figure 3

Install this device indoors only, and in any case do not use the pump in an environment exposed to atmospheric agents (rain, ice, snow), dust, aggressive gases, or in explosive environments or those with a high fire risk.

During operation, the following environmental conditions must be respected:

- maximum pressure: 2 bar above atmospheric pressure;
- temperature: from +5 °C to +35 °C;
- relative humidity: 0 – 90 % (non-condensing).

In the presence of magnetic fields the pump must be protected using a ferromagnetic shield. See the appendix "Technical Information" for detailed information.

The TwisTorr 74 FS pumps must be used in conjunction with one of the suitable Agilent controller and they must be connected to a primary pump (see "Technical Information").

Tab. 1

PART NUMBER	CONTROLLER
X3508-64301	TwisTorr 74 FS AG Rack RS232/485
X3508-64022	TwisTorr 74 FS AG Rack Profibus
X3509-64030	TwisTorr 74 FS on board 110/220V
X3509-64021	TwisTorr 74 FS on board 24V
X3510-64050	Controller TwisTorr 74 FS PCB

CAUTION! The TwisTorr 74 FS complies to installation category II as per directive EN 61010-1. Connect the device to a mains line that satisfies the above category.

For installation of optional accessories, see "Technical Information".

Pump Fixing

WARNING!



Failure to comply with these installation instructions could result in the pump detaching from the system in the event of a rotor failure, which could cause property damage or serious injury or death.

The TwisTorr 74 FS can be installed in any position. Fix the TwisTorr 74 FS in a stable position, mounting the inlet flange of the turbopump to the system counter-flange, with a connection capable of withstanding a torque of 250 Nm around its axis.

The following table shows, for each flange, the necessary number of clamps and the relevant fixing torque.

Tab. 2

FLANGE	FIXING DEVICE	N.	FIXING TORQUE
ISO 63 K	M10 clamps: IC63250DCMZ	4	22 Nm
CFF 4 ½	Mounting Kit X3502-68007 Metric screws X3502-68008 American screws X3502-68009	8	20 Nm
CFF 2 ¾	Screws	6	20 Nm
KF 40	Clamp KF40	1	-

The turbopump with ConFlat inlet flange must be fixed to the vacuum chamber by means of the appropriate Agilent hardware. See the appendix "Technical Information" for a detailed description.

NOTE

The TwisTorr 74 FS cannot be fixed by means of its base.

Use of TwisTorr 74 FS

WARNING!



Ensure the proper usage of the equipment.

The operator of this instrument is advised to use the equipment in a manner as specified in this manual.

If the guidance of this manual is not followed, the protection provided by the equipment may be impaired.

All the instructions to correctly use the turbopump are contained in the controller manual. Read carefully this manual before use the pump. To obtain better limit pressures it is possible to heat the pump.

While heating the vacuum chamber, the temperature of the inlet flange must not exceed 120 °C for a ConFlat flange and 80 °C for an ISO flange. While heating always use the water cooling.

WARNING!



Never use the turbopump when the inlet flange is not connected to the vacuum chamber. Do not touch the turbopump or any of its accessories during the heating process. The high temperatures may cause burns.

CAUTION!

Avoid impacts, oscillations or harsh movements of the pump when in operation. The bearings may become damaged. Use air or inert gas free from dust or particles for venting the pump. The pressure at the vent port must be less than 2 bar (above atmospheric pressure). For pumping gases containing particulate or aggressive pollutants for the bearings, these pumps are fitted with a special port (integrated purge/vent device) to allow a steady flow of inert gas (He, N₂, Ar) for pump bearing protection (see the annex "Technical Information").

CAUTION!

Never use the pump with corrosive gases or vapor to avoid damage to the internal materials of the pump.

WARNING!



When employing the pump for pumping toxic, flammable, or radioactive gases, please follow the required procedures for each gas disposal. Do not use the pump in presence of explosive gases. The pump is designed to pump high throughput of N₂, Ar and lighter gas. Should you need to pump gases heavier than Ar, please contact Agilent technical support for information.

Maintenance

The TwisTorr 74 FS does not require any maintenance. Any work must be carried out by authorized personnel.

WARNING!



Before carrying out any work on the system, disconnect it from the mains, vent the pump by opening the appropriate valve, wait until the rotor has stopped rotating and wait until the surface temperature of the pump falls below 50 °C.

In the case of breakdown, contact your local Agilent service center or "Agilent advanced exchange service" who can supply a reconditioned system to replace that broken down.

NOTE

Before returning the pump to the constructor for repairs, or advanced exchange service, the "Request for Return" sheet attached to this instruction manual must be filled-in and sent to the local sales office. A copy of the sheet must be inserted in the system package before shipping.

If a system is to be scrapped, it must be disposed of in accordance with the specific national standards.

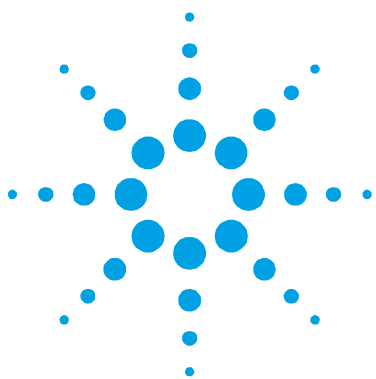
Disposal

Meaning of the "WEEE" logo found in labels. The following symbol is applied in accordance with the EC WEEE (Waste Electrical and Electronic Equipment) Directive. This symbol (**valid only in countries of the European Community**) indicates that the product it applies to must NOT be disposed of together with ordinary domestic or industrial waste but must be sent to a differentiated waste collection system. The end user is therefore invited to contact the supplier of the device, whether the Parent Company or a retailer, to initiate the collection and disposal process after checking the contractual terms and conditions of sale.



For more information refer to:

<http://www.agilent.com/environment/product/index.shtml>



17

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Description of the TwisTorr 74 FS

The TwisTorr 74 FS pump is available in various models that differ in the high vacuum flange.

The TwisTorr 74 FS pump models are:

- Model X3502-64170 with ISO 63 high vacuum flange;
- Model X3502-64171 with KF40 high vacuum flange;
- Model X3502-64172 with ConFlat 4.5" external diameter high vacuum flange;
- Model X3502-64173 with ConFlat 2.75" external diameter high vacuum flange;

The following figures show the 4 models.



Figure 4 Model X3502-64170

17 Technical Information
Description of the TwisTorr 74 FS



Figure 5 Model X3502-64171

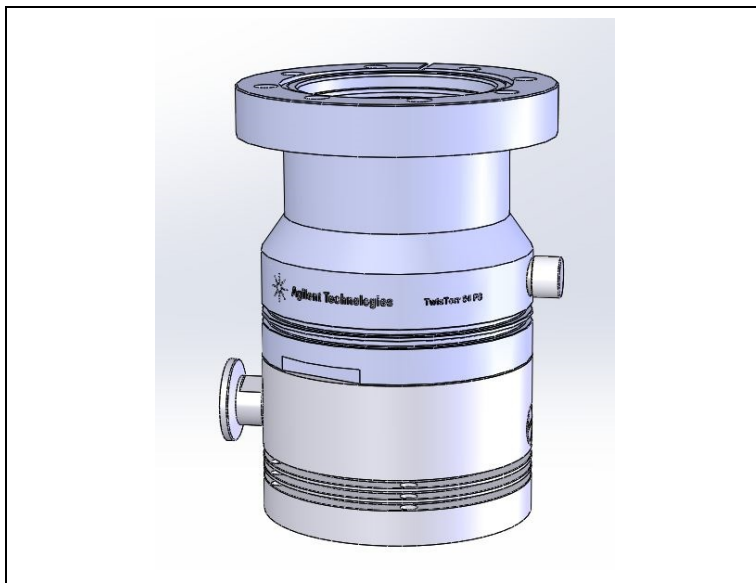


Figure 6 Model X3502-64172

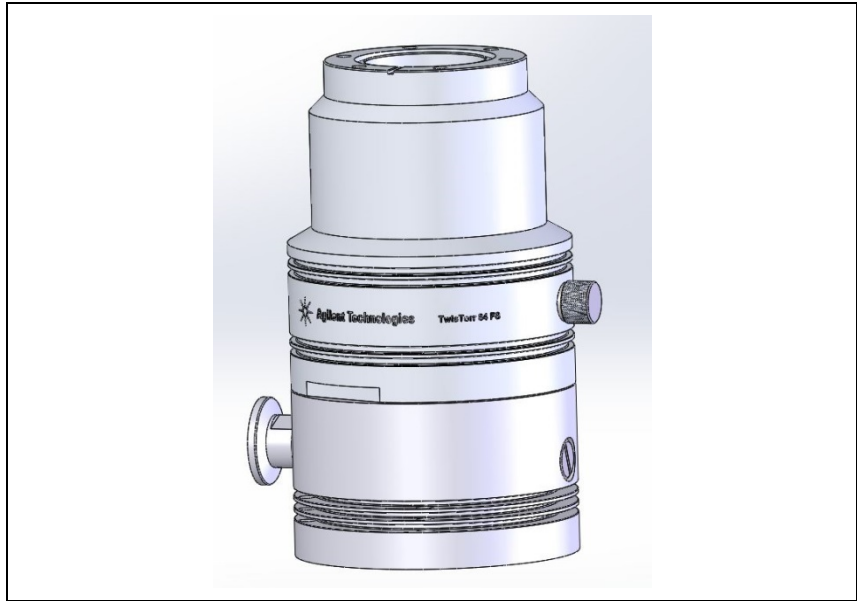


Figure 7 Model X3502-64173

Pump Description

The TwisTorr 74 pump consists of a high frequency motor driving a turbine fitted with 9 bladed stages and 3 TwisTorr drag stages. The turbine rotates in an anticlockwise direction when viewed from the high vacuum flange end.

The turbine is made of high-strength aluminium alloy, machined from a single block. The TwisTorr stages are in the form of four disks.

The turbine rotor is supported by permanently lubricated high precision ceramic ball bearings installed on the forevacuum side of the pump.

The static blades of the stator are made of stainless steel. These are supported and accurately positioned by spacer rings.

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Description of the TwisTorr 74 FS

The TwisTorr stators are in the form of selfpositioning machined disks. On each disk there are parallel spiral pumping channels designed to pump in centrifugal direction on one side and in centripetal direction on the other side. These are made of aluminium alloy.

During normal operation, the motor is fed with a voltage of 54 Vac three-phase at 1167 Hz. To reduce losses during start-up to a minimum, the frequency increases according to a ramp with a higher initial voltage/frequency ratio.

WARNING!

The pump must always be cooled before starting using it. Running the pump without cooling it could damage inside components.

The pump can be water cooled or air cooled: in the first case an external optional water cooling kit is available (model X3502-68002 , X3502-68003), in the second case an external optional fan is available (model 9699290). A thermistor sensor is mounted near the upper bearing to prevent the pump from overheating.

If the thermistor reading exceeds 60 °C the pump fails and stops with Error Code = 2. If the thermistor reading is below 0 °C, the pump fails and stops with Error Code = 1 (see Controller Manual),

The pump can operate in any position and can be supported on the high vacuum flange. The connection of the forevacuum on the side of the pump is a KF 16 NW flange.

Technical Specification

Max gas throughput:

- water cooling (ambient temperature=35°C; water temperature= 25°C; backing pump 5 m3/h)
- air cooling (ambient temperature=35°C; backing pump 5 m3/h)

Tab. 3 Technical Specification

Pumping speed	ISO 63 / CFF 4.5"	CFF 2.75"	KF 40
N ₂	60 l/s	50 l/s	44 l/s
He	54 l/s	40 l/s	33 l/s
H ₂	42 l/s	31 l/s	28 l/s
Ar	59 l/s	51 l/s	40 l/s
Max gas throughput at full rotational speed (with recommended forepump)	Air cooling		Water cooling
N ₂	130 sccm		130 sccm
Ar	80 sccm		80 sccm
Compression ratio			
N ₂	1×10^9		
He	2×10^5		
H ₂	1×10^4		
Ar	$> 1 \times 10^9$		
Foreline Tolerance(*)			
N ₂	> 12 mbar		
He	> 10 mbar		
H ₂	> 4 mbar		
Ar	> 14 mbar		
Base pressure with recommended forepump	$< 5 \times 10^{-10}$ mbar $(< 3.75 \times 10^{-10}$ Torr) (According to standard DIN 28 428, the base pressure is that measured in a leak-free test dome, 48 hours after the completion of test dome bake-out, with a Turbopump fitted with a ConFlat flange and using the recommended pre-vacuum pump)		

(*) The Foreline Tolerance is the pressure at which the turbopump still produce a compression of 100

17 Technical Information

Technical Specification

Inlet flange	CFF 4.5" O.D. CFF 2.75" O.D.	ISO 63 KF 40
Foreline flange	KF16 NW	
Rotational speed	70000 rpm (1167 Hz driving frequency)	
Start-up time	< 2 minutes	
Recommended forepump	mechanical: Agilent DS 40M mechanical: Agilent DS 102 dry pump: Agilent IDP-3 dry pump: Agilent IDP-7	
Operating position	Any	
Operating ambient temperature	+5 °C to +35 °C	
Relative humidity of air	0 – 90 % (not condensing)	
Bakeout temperature	80 °C at inlet flange (ISO63) and 120°C (CFF 4.5") max	
Lubricant	Permanent lubrication	
Cooling requirements	Max forced cooling air temperature = 35°C Max cooling water temperature = 25°C	
Coolant water	Minimum flow: 65 l/h (0.30 GPM) Temperature: +15 °C to +25 °C Pressure: 2 to 4 bar (30 to 60 psi)	
Noise Pressure level (pump only)	40 ± 4 dB(A) at 1 meter † (Min = 32 dB(A), Max = 48 dB(A))	
Pwr supply:		
Input Voltage:	76 Vac, 3 Phase (provided by the controller)	
Input Frequency:	1167 Hz max (provided by the controller)	
Max input pwr:	130 VA	
Stand-by pwr:	8 W	
Max oper. power:	70 W (according to controller model) with water or air cooling	
Installation category	II	
Pollution degree	2	
Storage temperature	-40° C to +70° C	
Max operating altitude	3000 m	

† average value ± standard deviation.

Weight kg (lbs):

Pump ISO 63 2.05 kg

Pump CFF 4.5'' 3.50 Kg

Pump KF 40 2.37 kg

Pump CFF 2.75'' 3.34 kg

TwisTorr 74 FS Outline

The following figures show the TwisTorr 74 FS (dimensions are in mm [inches]).

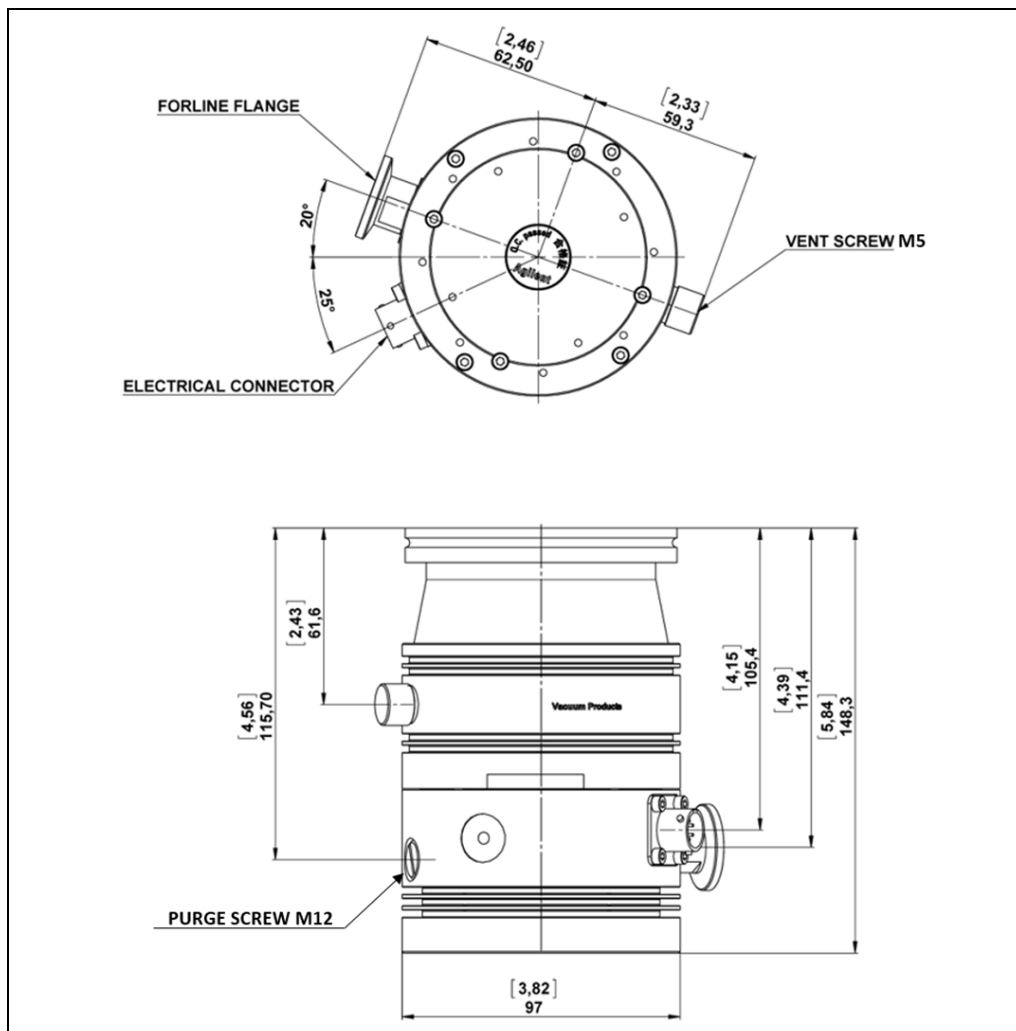


Figure 8 TwisTorr 74 FS ISO 63 (X3502-64170)

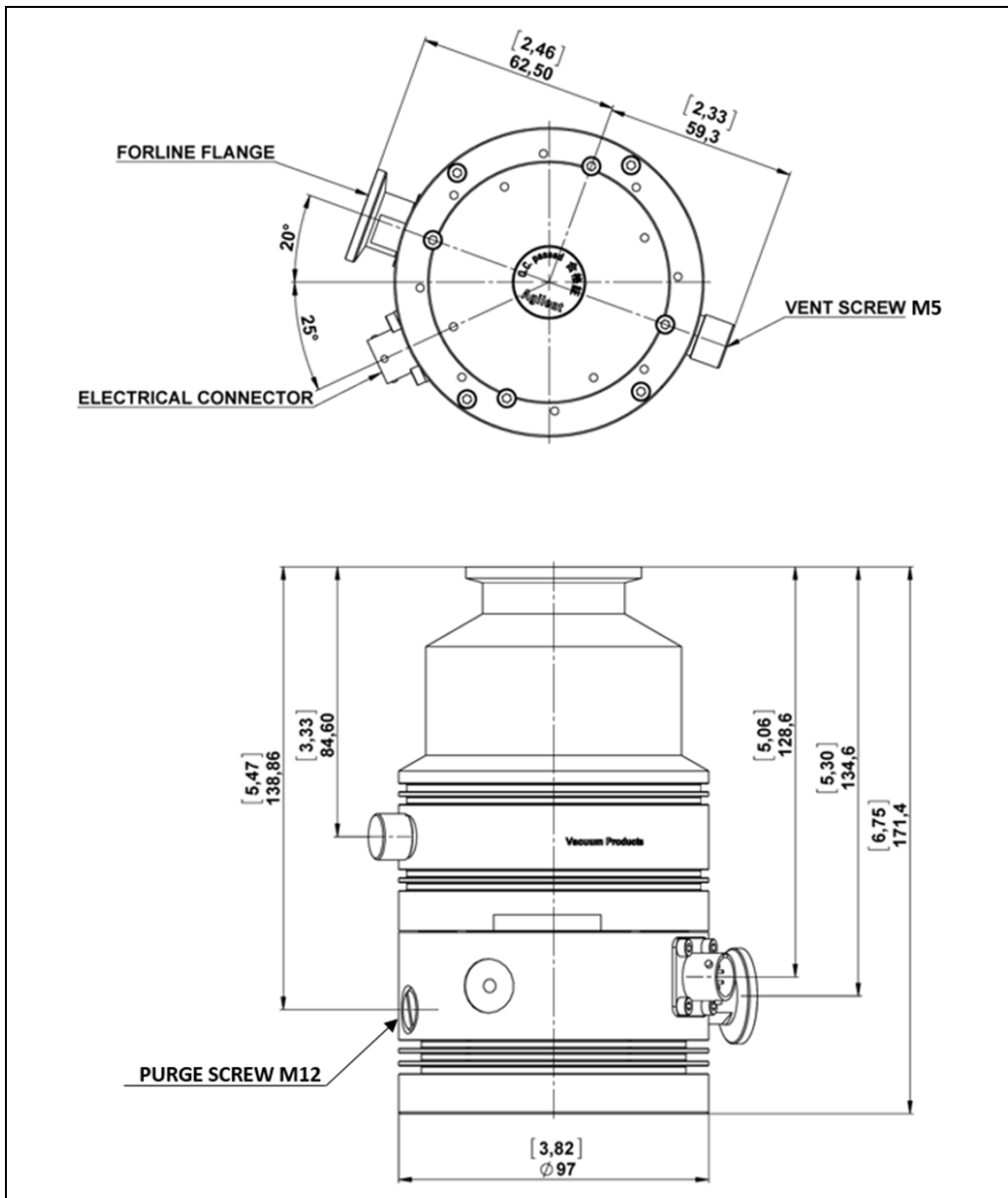


Figure 9 TwisTorr 74 FS KF 40 (X3502-64171)

17 Technical Information
TwisTorr 74 FS Outline

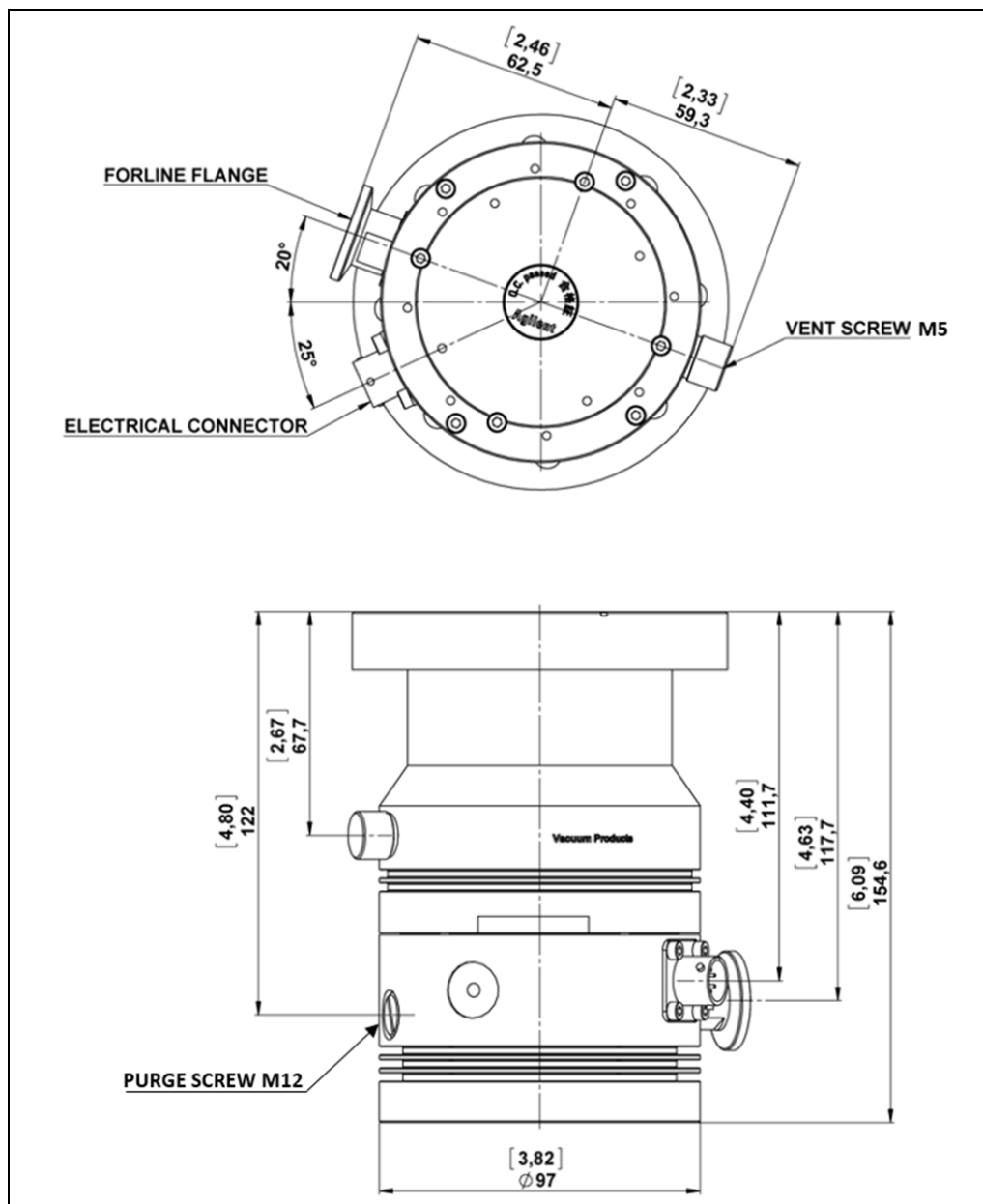


Figure 10 TwisTorr 74 FS CFF 4.5'' (X3502-64172)

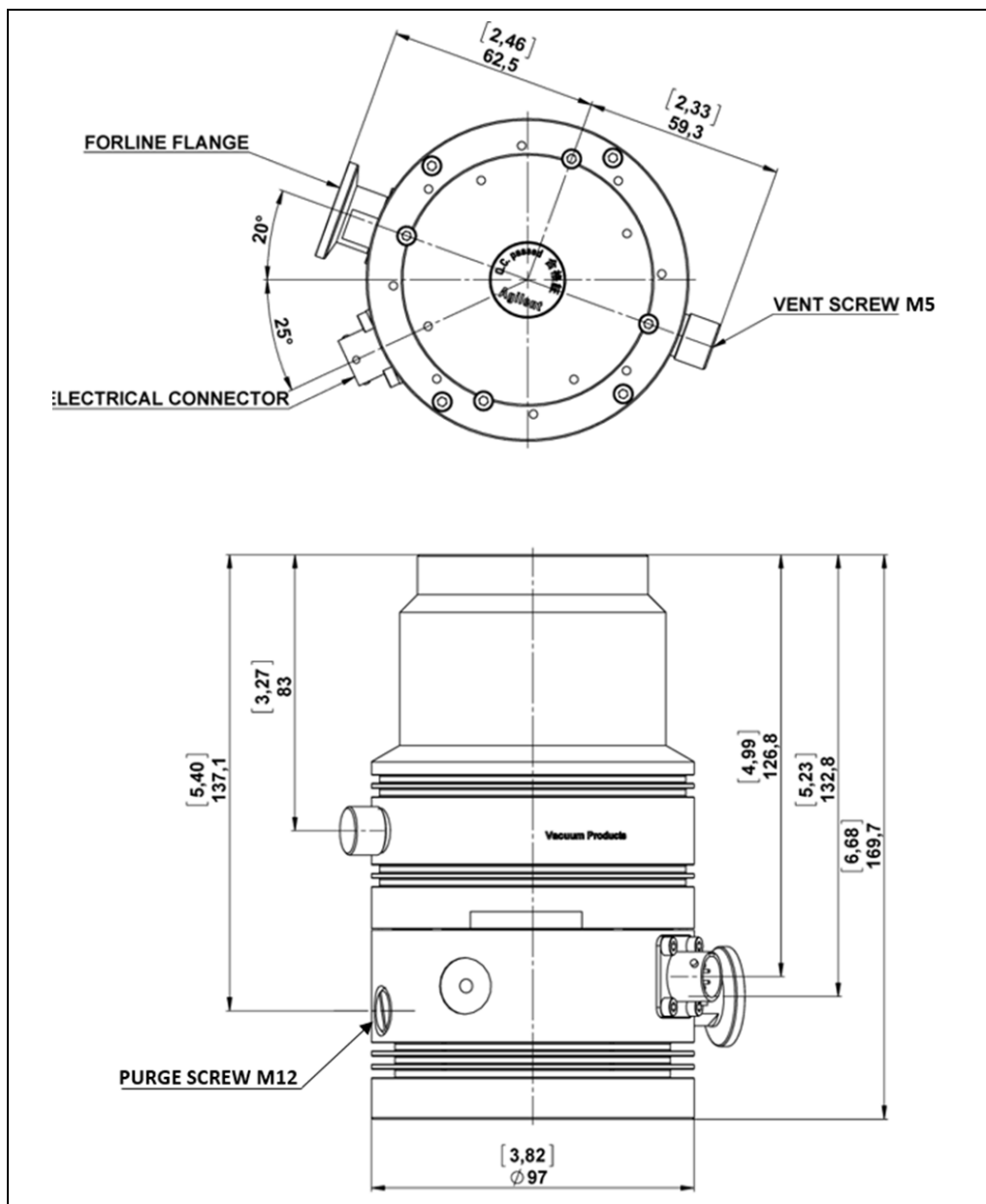


Figure 11 TwisTorr 74 FS CFF 2.75'' (X3502-64173)

17 Technical Information

TwisTorr 74 FS Outline

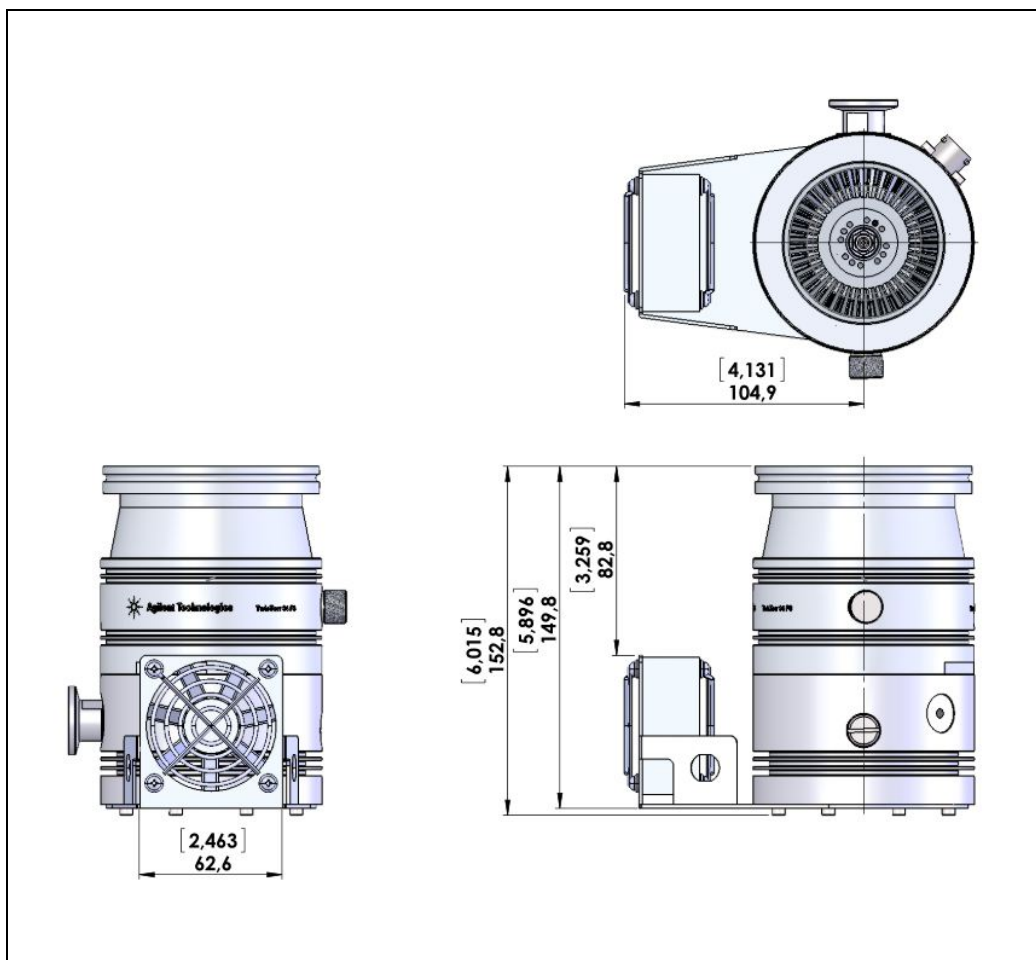


Figure 12 TwisTorr 74 FS ISO 63 (X3502-64170) with air cooling kit

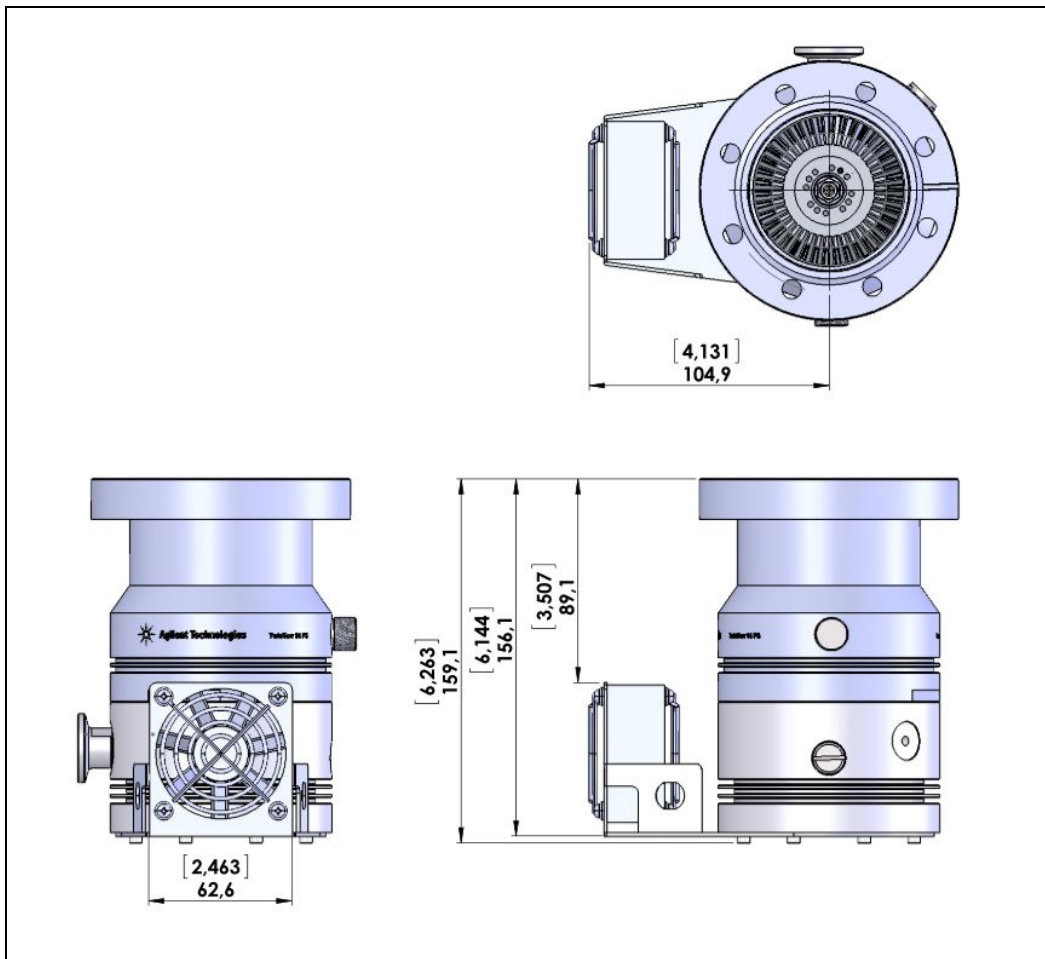


Figure 13 TwisTorr 74 FS CFF 4 1/2 (X3502-64172) with air cooling kit

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TwisTorr 74 FS Outline

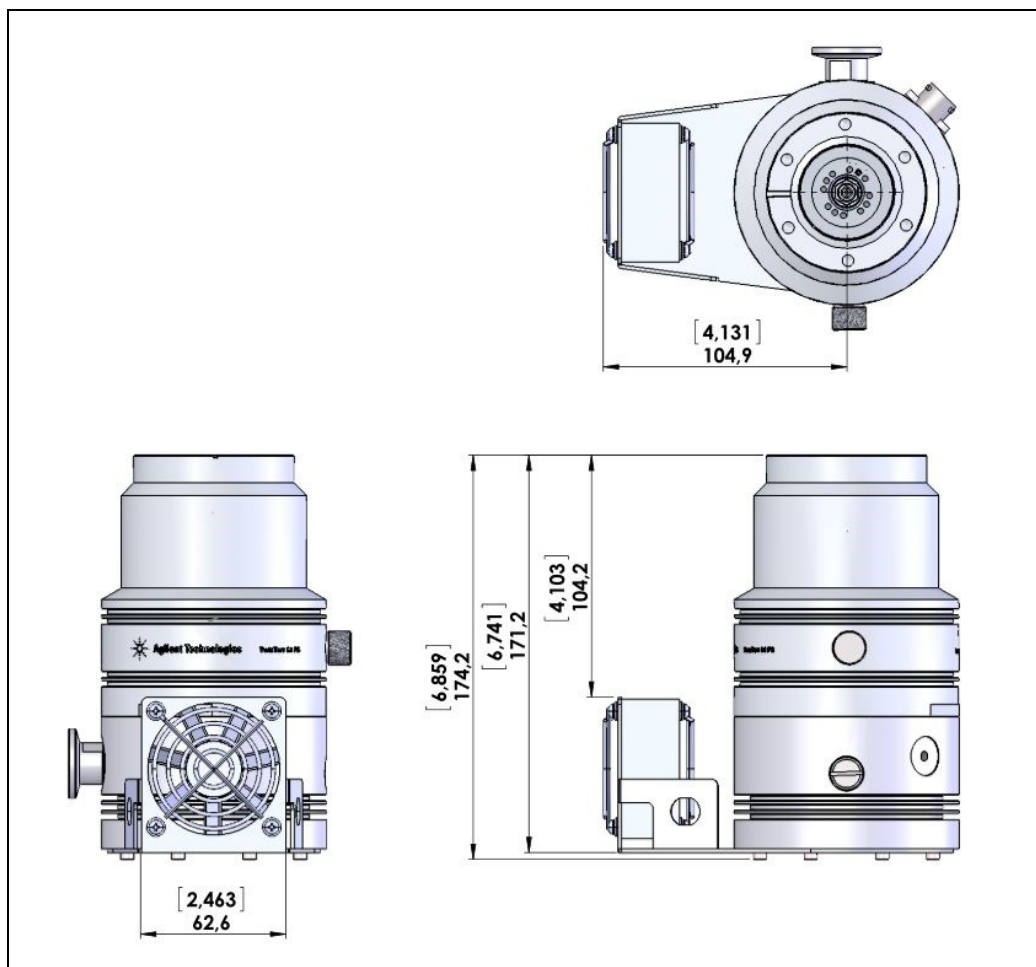


Figure 14 TwisTorr 74 FS CFF 2.75'' (X3502-64173) with air cooling kit

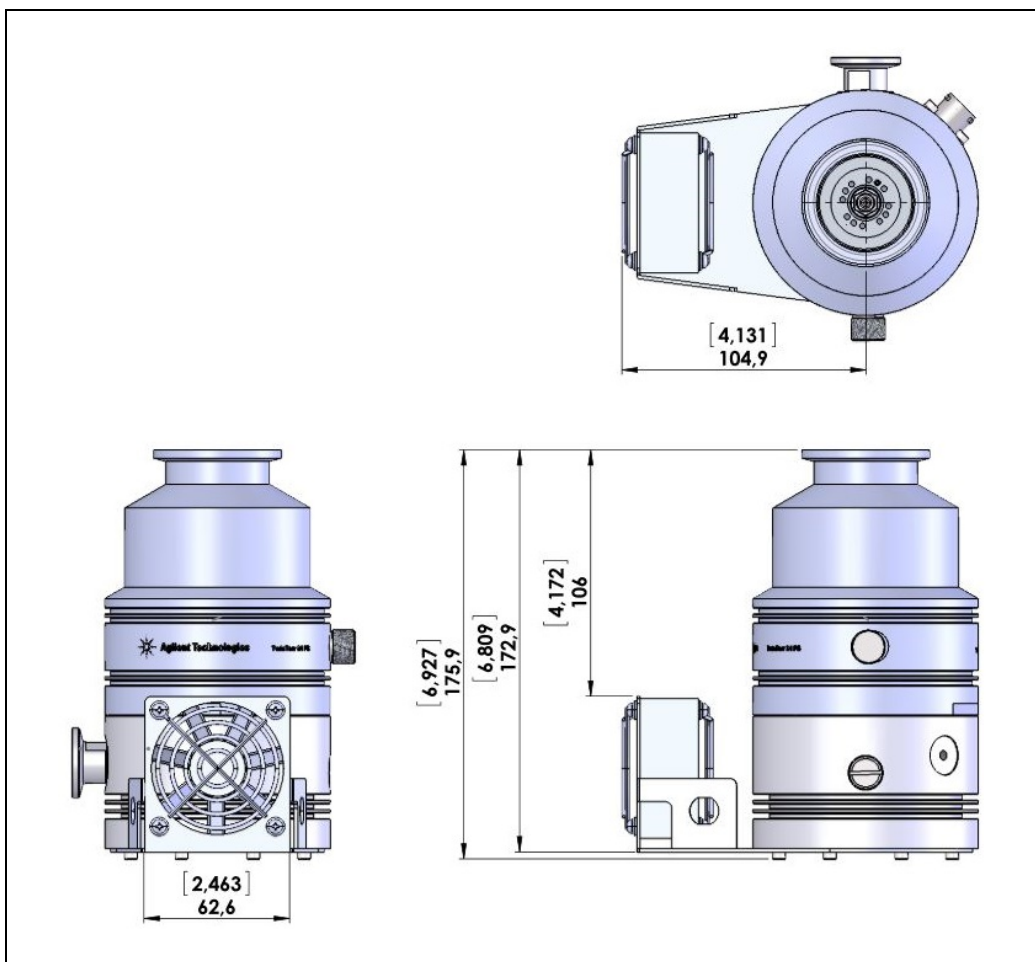


Figure 15 TwisTorr 74 FS KF 40 (X3502-64171) with air cooling kit

17 Technical Information
TwisTorr 74 FS Outline

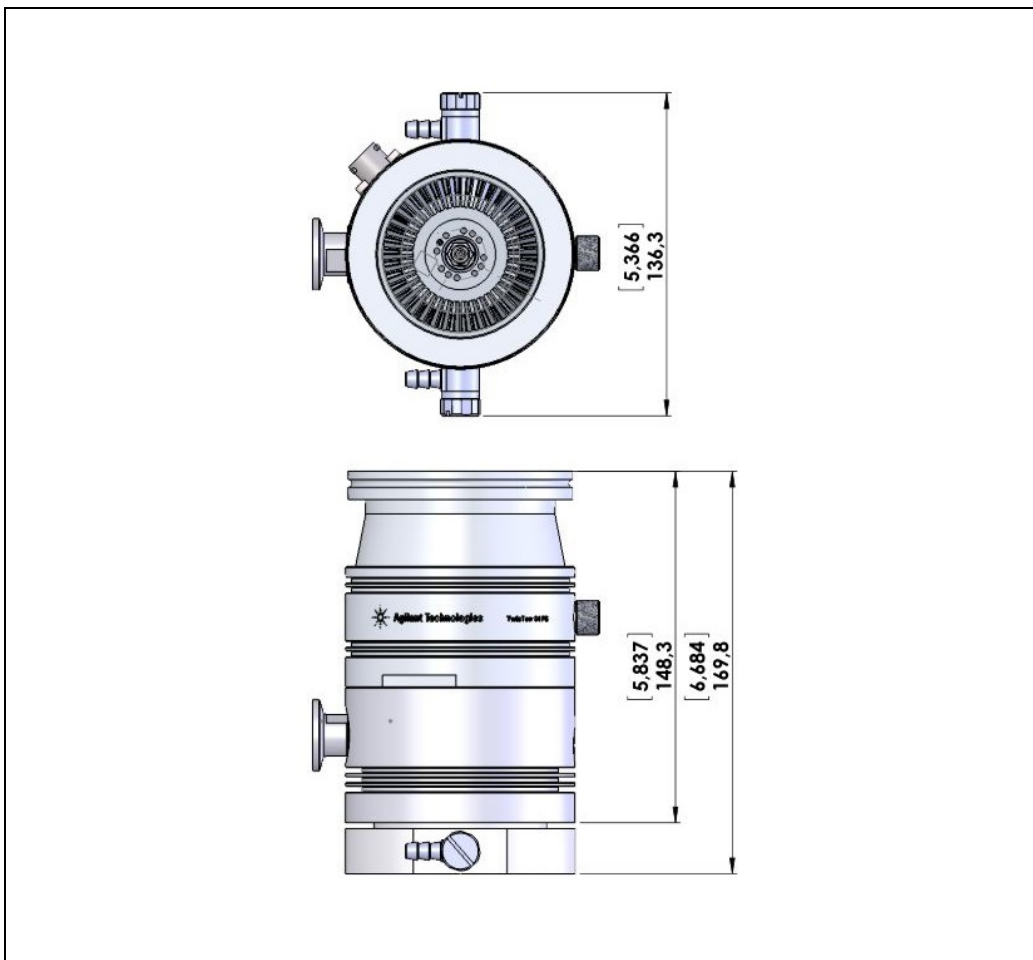


Figure 16 TwisTorr 74 FS ISO 63 (X3502-64170) with water cooling kit

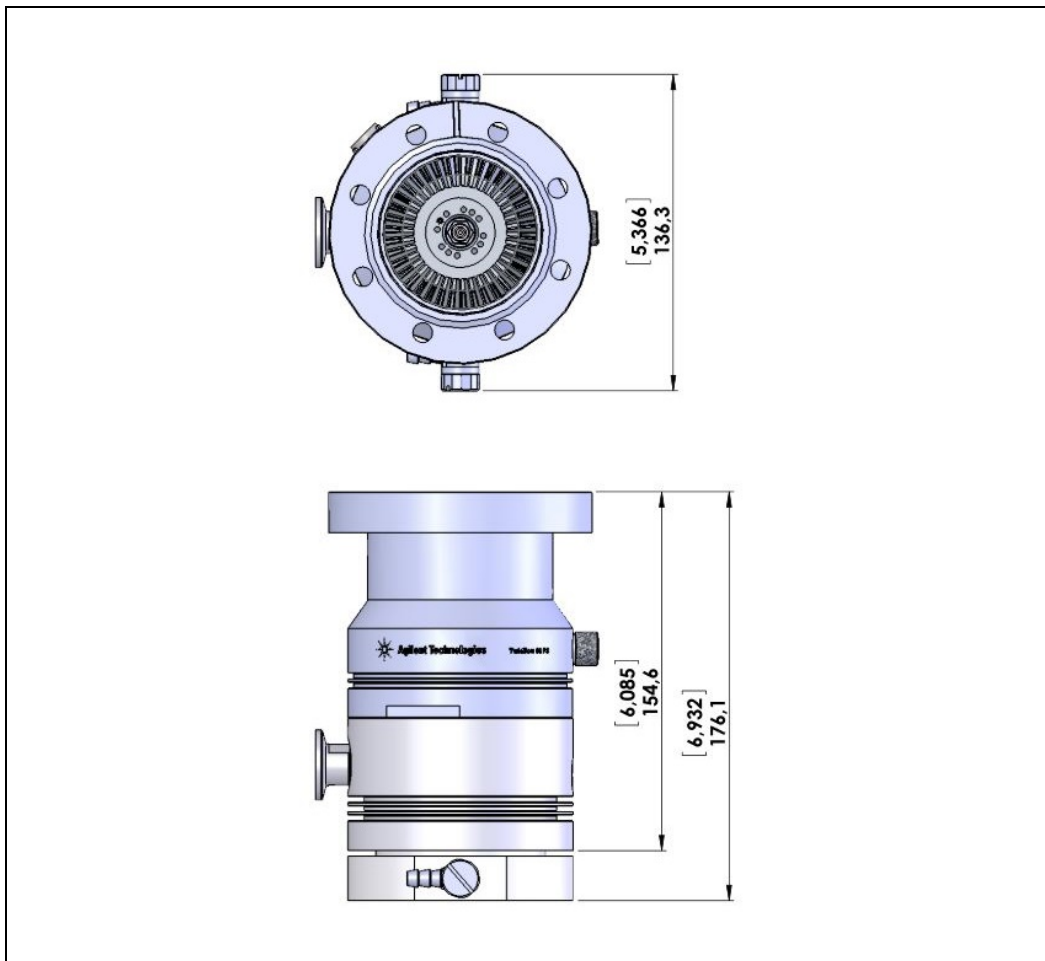


Figure 17 TwisTorr 74 FS CFF 4 1/2 (X3502-64172) with water cooling kit

17 Technical Information

TwisTorr 74 FS Outline

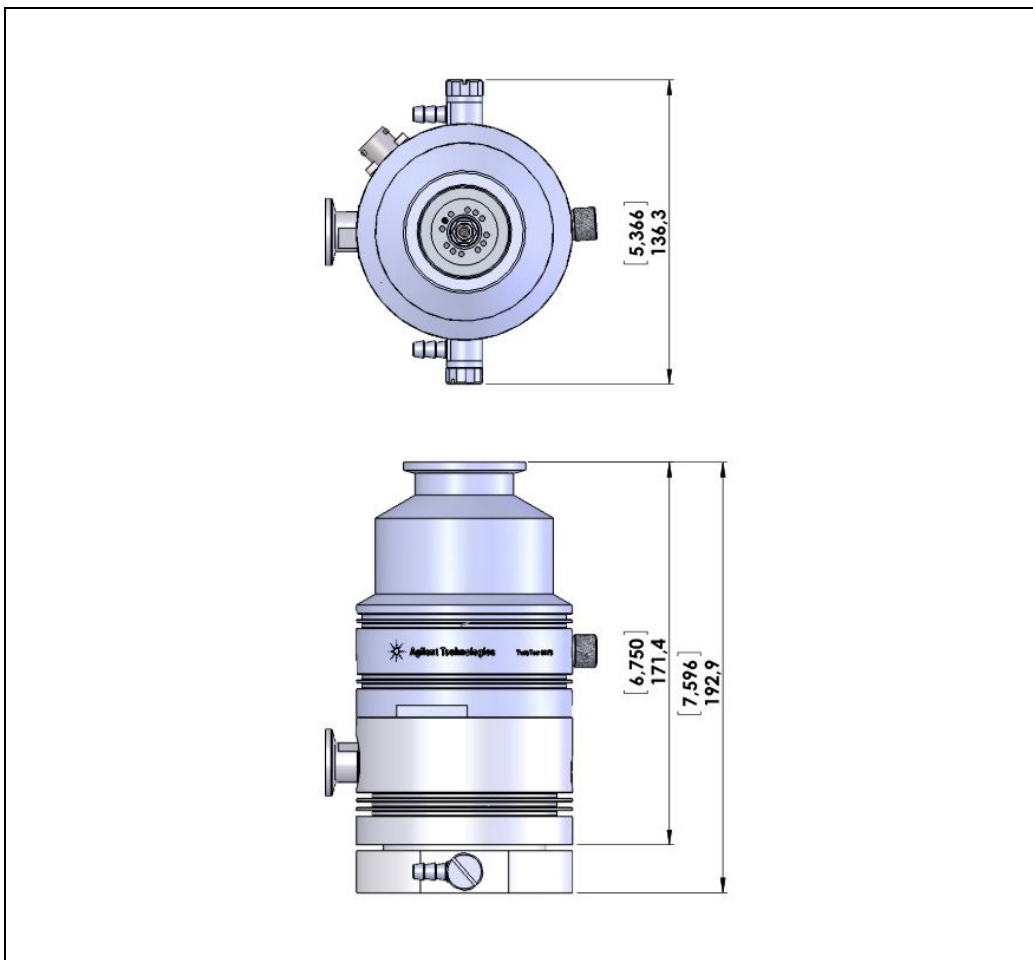


Figure 18 TwisTorr 74 FS CFF 2.75'' (X3502-64173) with water cooling kit

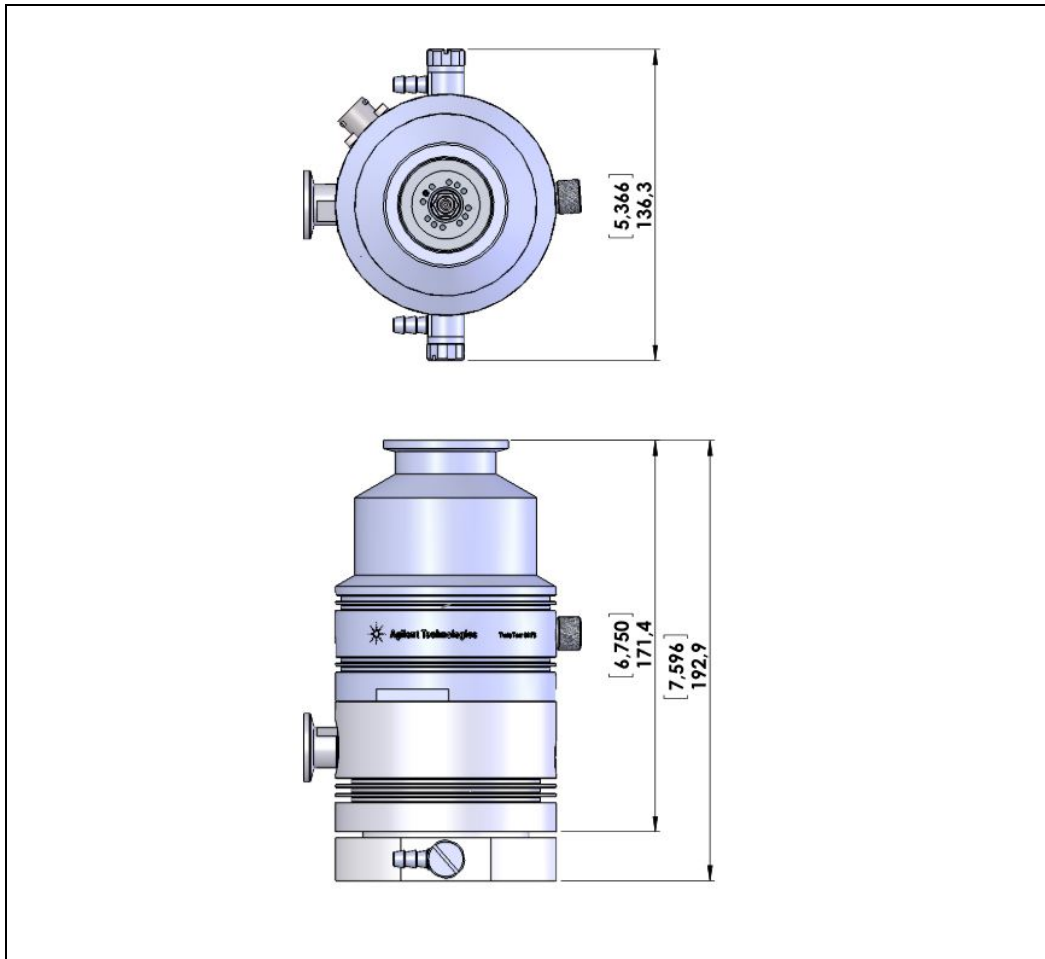


Figure 19 TwisTorr 74 FS KF 40 (X3502-64171) with water cooling kit

17 Technical Information

TwisTorr 74 FS Outline

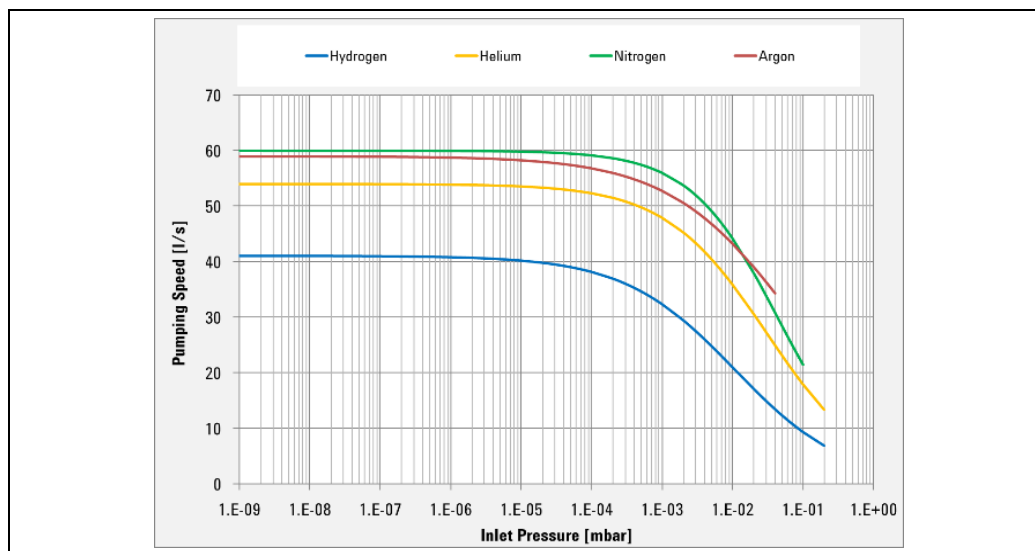


Figure 20 Graph of Pumping Speed vs. Inlet Pressure for TwisTorr 74 FS pump

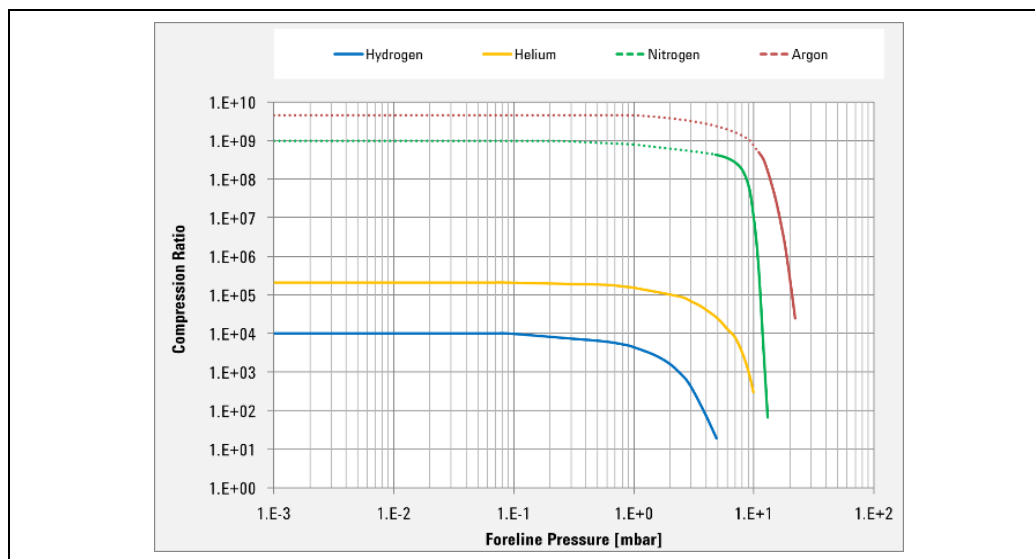


Figure 21 Graph of Compression ratio vs. Foreline Pressure for TwisTorr 74 FS pump

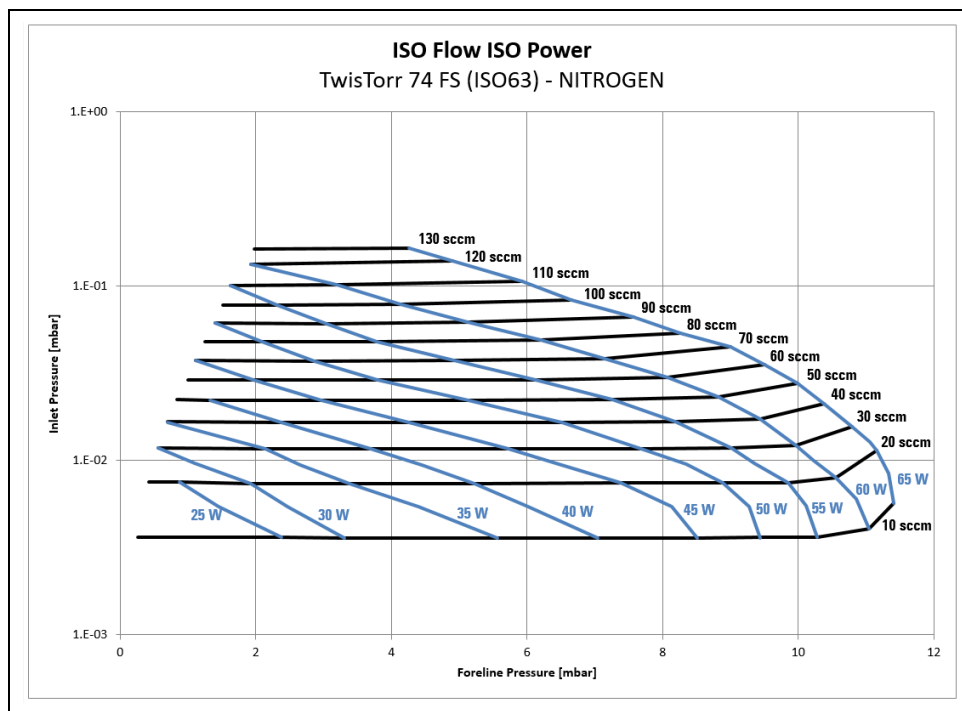


Figure 22 Graph of Iso Flow – Iso Power TwisTorr 74 FS pump: Nitrogen

17 Technical Information

TwisTorr 74 FS Outline

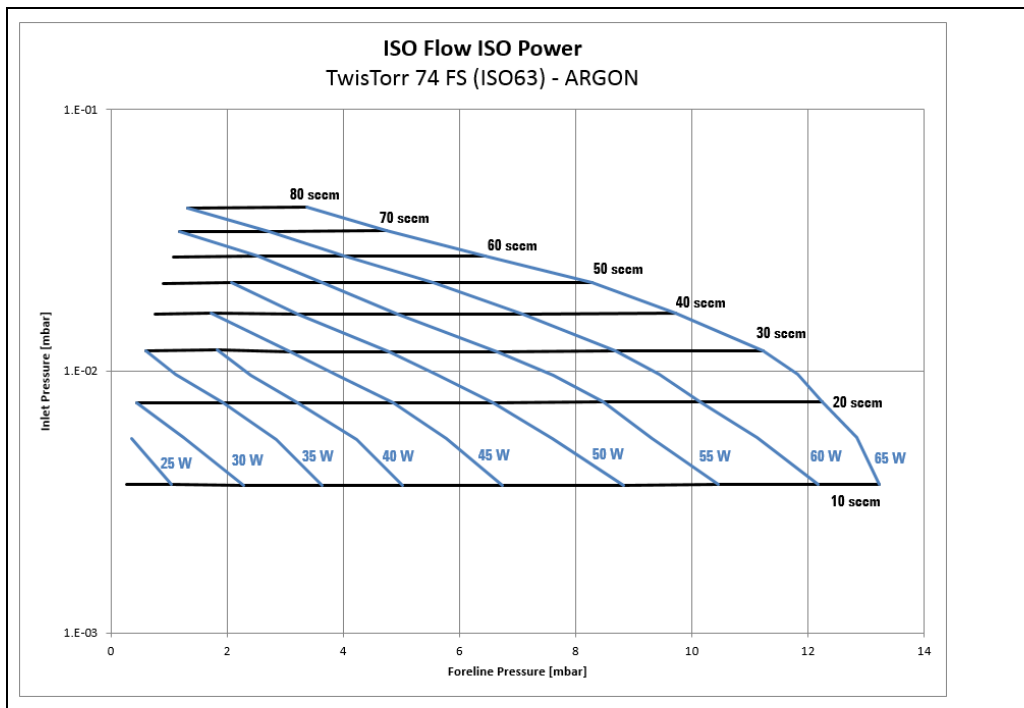


Figure 23 Graph of Iso Flow – Iso Power TwisTorr 74 FS pump: Argon

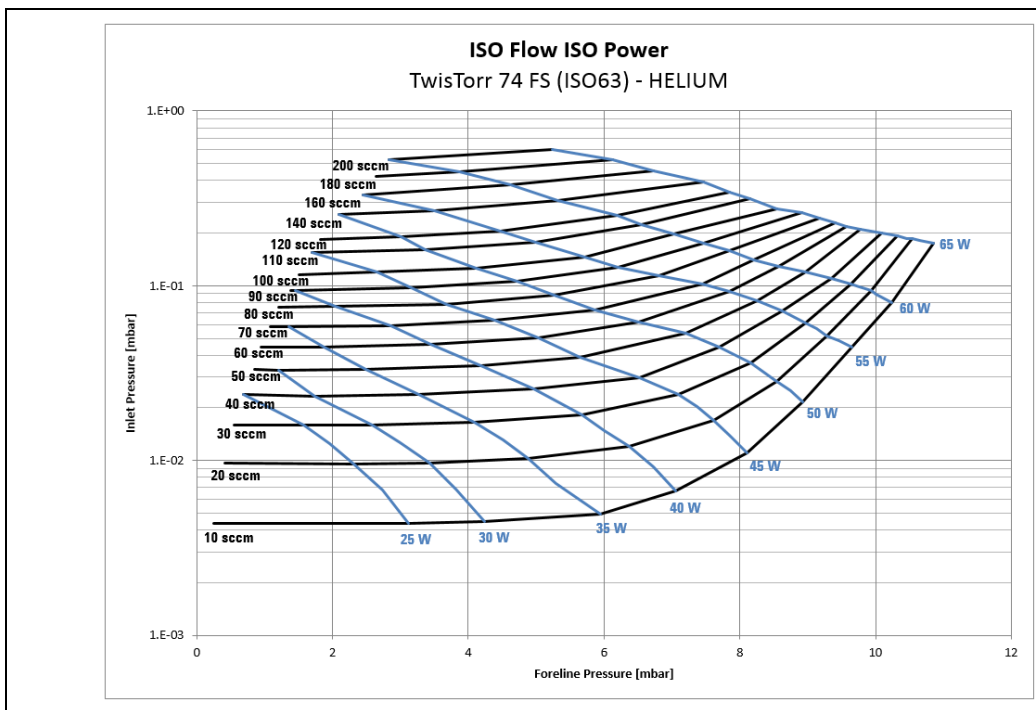


Figure 24 Graph of Iso Flow – Iso Power TwisTorr 74 FS pump: Helium

17 Technical Information

TwisTorr 74 FS Outline

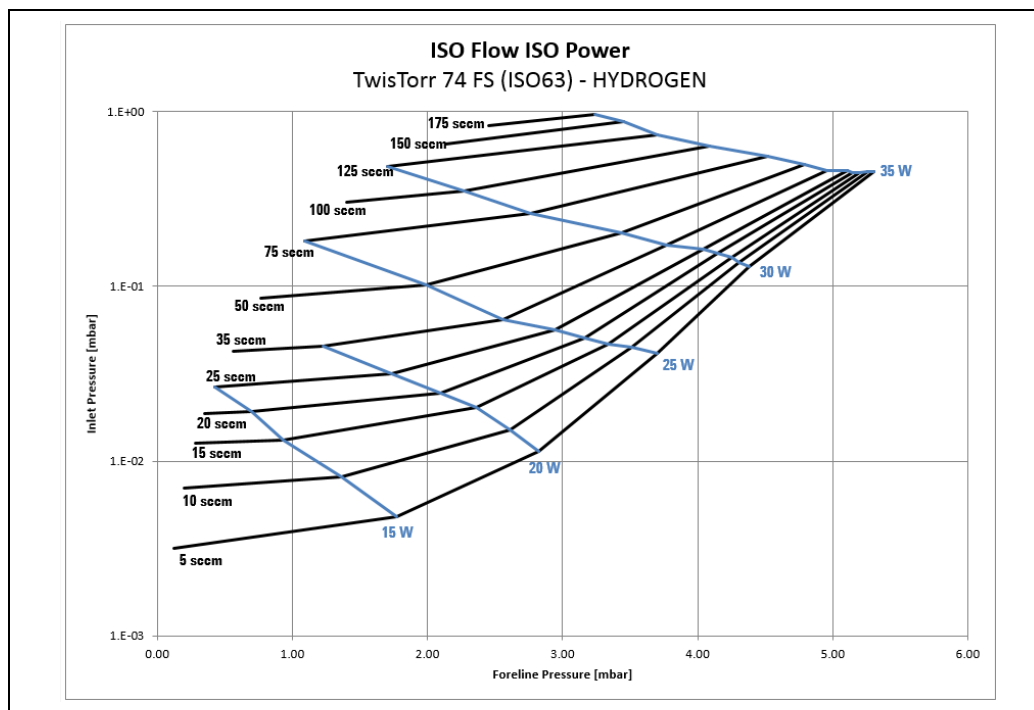


Figure 25 Graph of Iso Flow TwisTorr 74 FS pump: Hydrogen

Interconnections

The following figure shows the TwisTorr 74 FS interconnections.

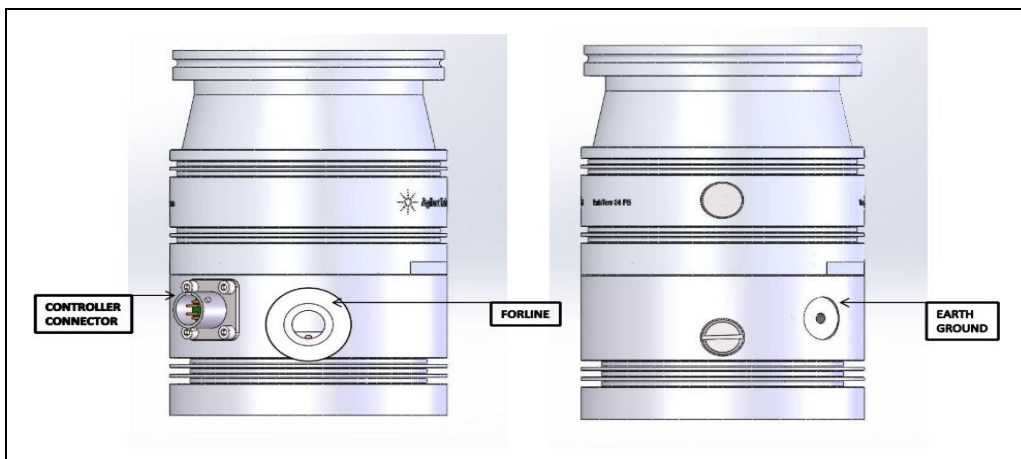


Figure 26

Earth (Ground) Connection

It is required to connect the pump body to protective earth with a M5 screw (marked thread in the pump body) using a green/yellow wire of size AWG 16. Please ensure a reliable connection with using an M5 screw and a snap ring or washer disc of the same metal as the pump body to prevent corrosion. The screw must be fixed with a 2 Nm torque.

The impedance between the pump-body and the earth connection point must be $< 0.1 \Omega$.

Connection C – Electrical

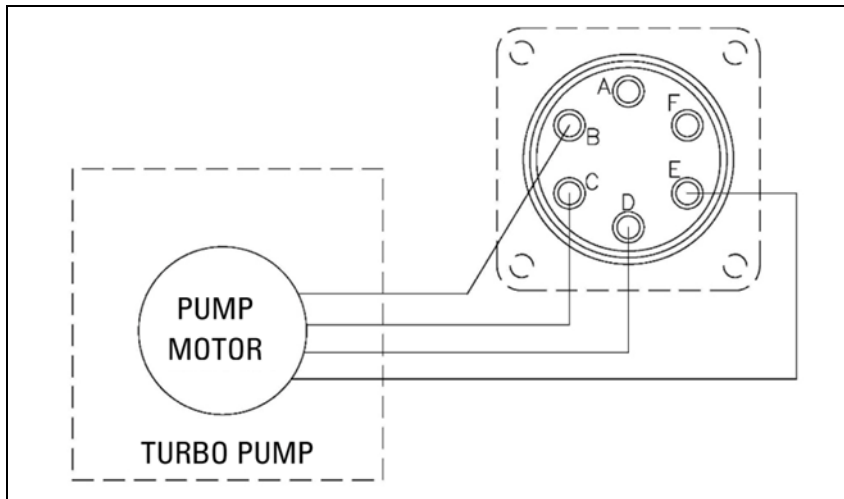


Figure 27

The turbopump is connected to the controller through a 6-pin connector. Pins B, C and D are the 3-phase supply to the motor, pin E is connected to the pump ground, pins A and F reserved for Agilent.

If the temperature sensor is disconnected, the pump will not start. To prevent damage to the pump when the temperature exceeds 60 °C, the sensor automatically cuts off the power supply.

Soft Start

To start the pump after a protracted stop of the TwisTorr 74 FS, it's recommended to enable the "Soft Start" function; this allows a better grease distribution in the bearings.

The "Soft Start" mode is disabled by default.

To enable the "Soft Start" mode it must be activated by the suitable software or Remote I/O command (see Controller Manual).

"Soft Start" frequency steps are the following: 450Hz, 1100Hz, 1167Hz or more details see Controller Manual.

Inlet screen installation

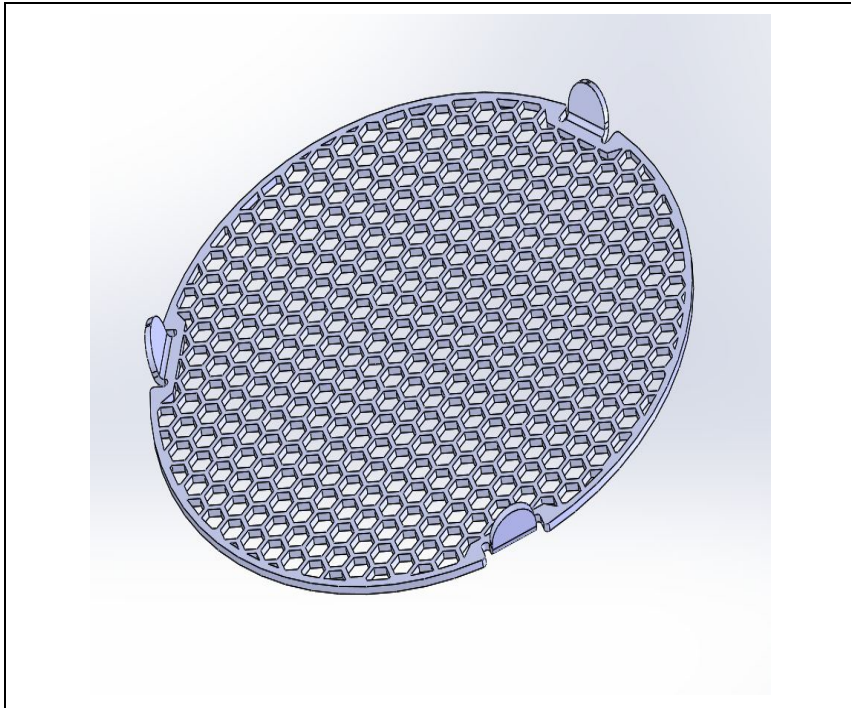


Figure 28 Model X3502-68000

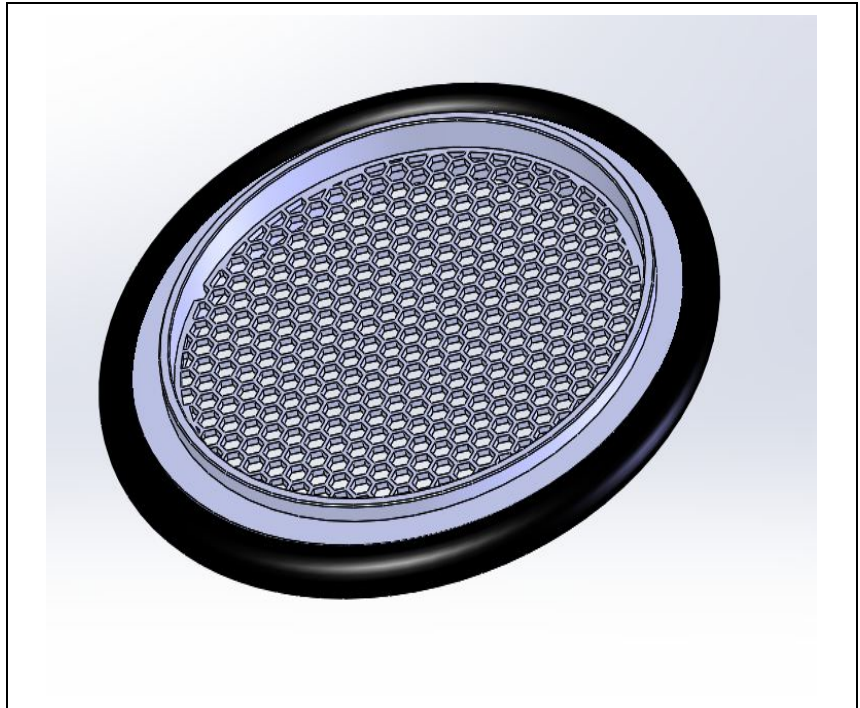


Figure 29 Model X3502-68001

The inlet screen models are the following;

- X3502-68001 (ISO 63)
- X3502-68000 (CFF 4.5’')
- 9699309 (KF 40)
- 9699328 (CFF 2.75’')

17 Technical Information

Inlet screen installation

prevent the rotor blades from being damaged by debris. The inlet screen, however, does reduce the pumping speed as per following table:

Reduction in pumping speed (%)				
	Ar	N ₂	He	H ₂
X3502-68000	20%	18%	7%	5%
X3502-68001				

The inlet screen is fitted in the upper part of the pump, as shown in the figure.



Figure 30

The screen can be mounted on each pump.

The screen can be removed as shown in the following figure.



Figure 31

The overall flange dimensions with the protection screen fitted on pump do not change as the inlet screen remains integrated into the center-ring.

Air Cooling Kit Installation

An air cooling kit (9699290) is available to improve the TwisTorr 74 FS cooling during operational conditions.

Fan specifications:

- air flow: 12.8 l/s (27.1 CFM)
- input voltage: 24 Vdc
- dimensions: 60 x 60 x 25 mm (2.36 x 2.36 x 0.98 in.)
- power: 2.6 W

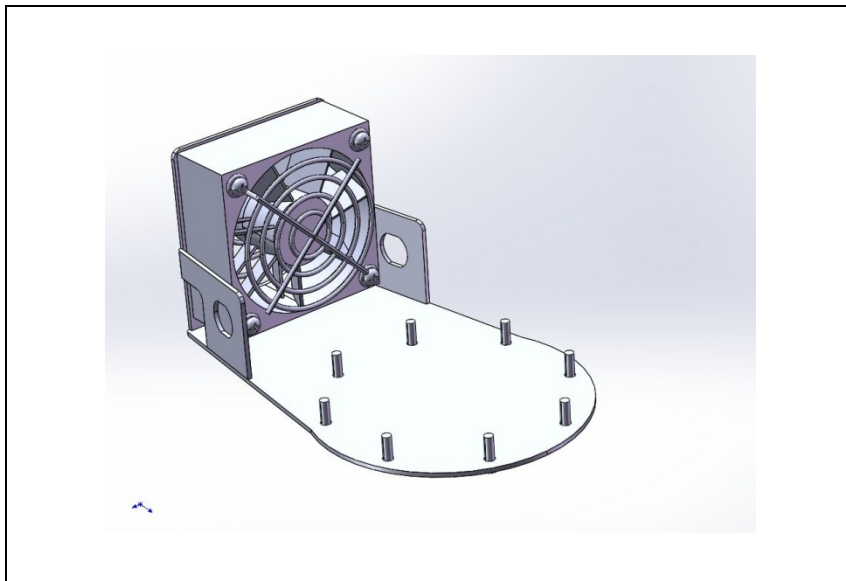


Figure 32

The fan bracket is shaped so that it can be mounted on the pump bottom (fixing thread = M3, diameter = 80,75 mm). When the pump is powered by the rack C.U., the use of the fan extension cable 9699940 is recommended.

To fix the fan to the TwisTorr 74 FS pump bottom, proceed as follows (see figure below):

- 1** Fix the fan to the suitable bracket by means of the furnished screws;
- 2** Fix the bracket to the pump body between the pump and the controller;
- 3** Connect the fan supply to the P4 connector of the controller.

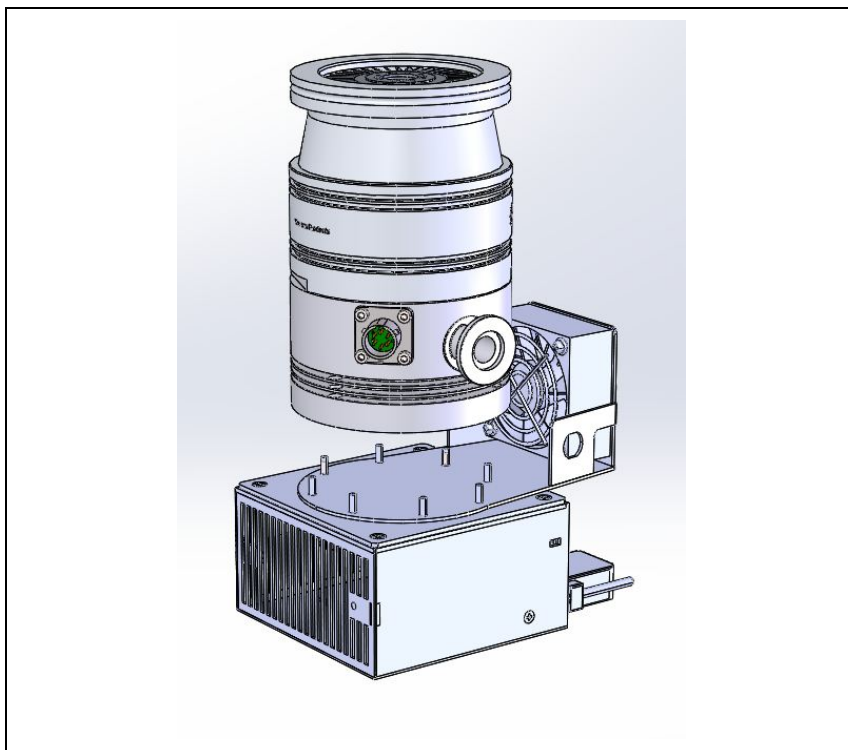


Figure 33

Water Cooling Kit Connection

A water cooling kit is available to be mounted when the pump is used under heavy load conditions or when air cooling is insufficient.

The model part numbers are:

- X3502-68002 (Metal),
- X3502-68003 (Plastic).

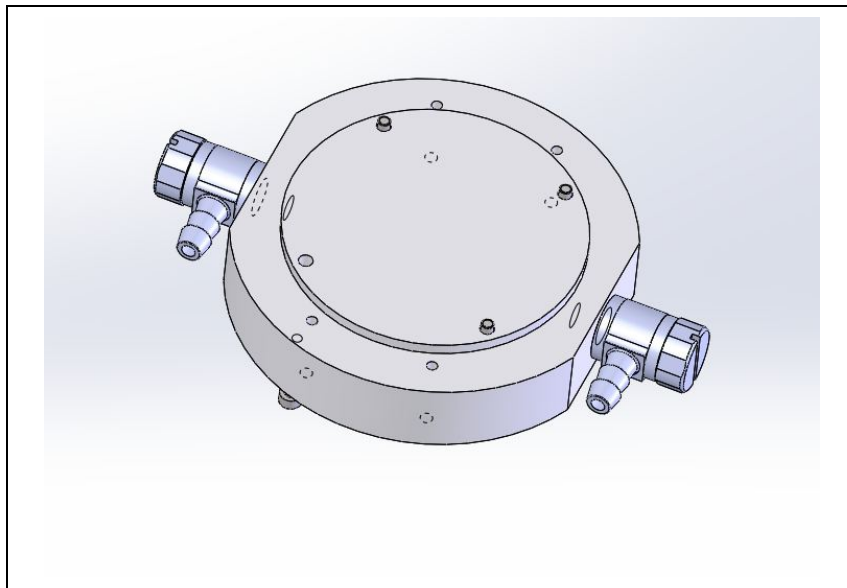


Figure 34 Model X3502-68002

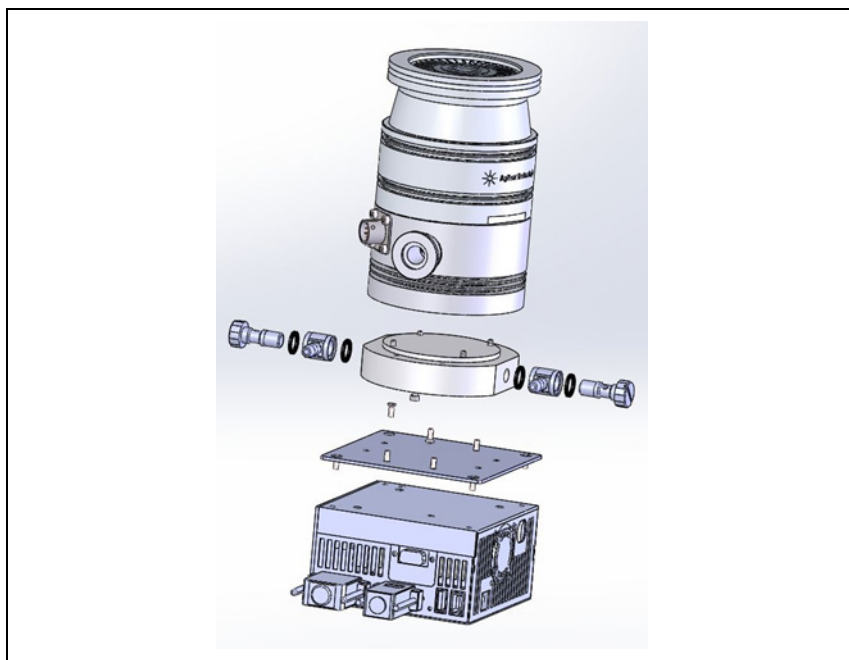


Figure 35 Model X3502-68002

CAUTION!

The items of the plastic model kit must be assembled as shown in the following figure

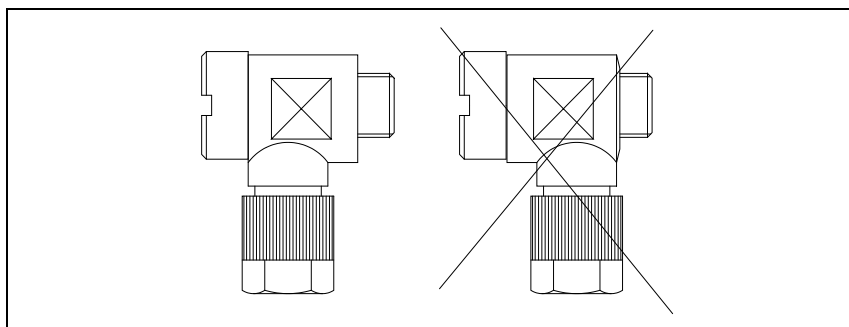


Figure 36

17 Technical Information

Water Cooling Kit Connection

The assembled kit must be screwed into the suitable holes of the pump body with a recommended closing torque of 5 Nm.

Cooling may be carried out either through an open circuit with eventual discharge of the water, or using a closed circuit cooling system.

The water temperature must be between +15 °C and +25 °C, with an inlet pressure between 2 and 4 bar. This allows a flow higher than 65 l/h.

NOTE

The water electrical conductance must be $\leq 500 \mu\text{s/cm}$. When the conductance is higher, in closed water circuit, the use of up to 20 % of Ethyl-Glycole is suggested.

Vent Accessories

The vent valve (9699844) allows to avoid undesired venting of the pump during temporary power failure and enables an automatic vent operation.

This vent valve waits before opening a minimum time of about 5 sec. This time can be increased up to about 220 min. by means of a setting of the controller software (optional).

To install the vent valve, remove the M5 threaded plug (see figure below).



Figure 37

Then screw the vent valve into the pump and tighten it using a 16 mm hexagonal spanner with a torque of 2.5 Nm.

17 Technical Information

Vent Accessories

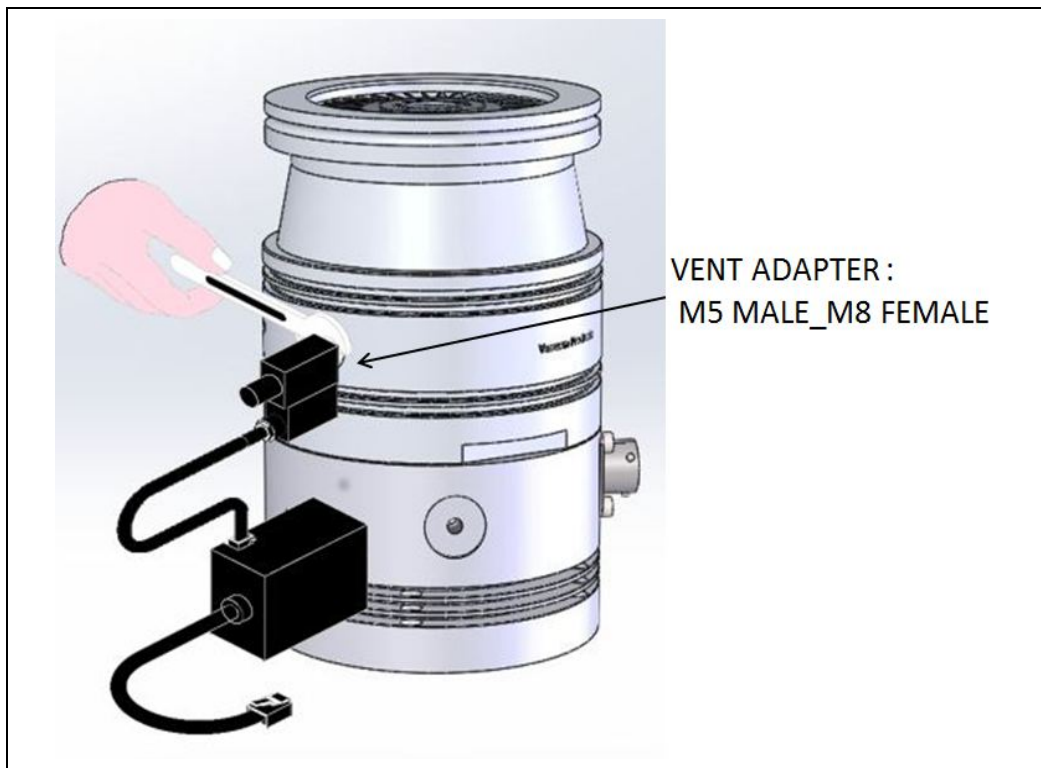


Figure 38

CAUTION!

Do not overtighten the valve as this may damage the thread on the pump.

Then connect the cable from the valve to the suitable connector on the controller.

Vibration Isolator Installation

Two vibration isolators for ISO and CFF inlet flange version pumps are available as accessories.

The two model part numbers are the following:

- model 9699375 for ISO 63 flange;
- model 9699376 for CFF 4.5" flange.

They typically reduce the vibration transmitted from the TwisTorr 74 FS to the system by a factor of 20.

Please refer to the relevant instruction manual.

Purge Valve Installation

A gas purge valve is available to protect the pump bearings against particulate and corrosive gases that could move into the pump. To install the gas purge valve it is necessary to unscrew the M12 screw from the purge port as shown in the following figure;



Figure 39

and then screw the M12 gas purge valve (with a torque of 2.5 Nm).

TwisTorr 74 FS On Board Controller Installation

The on board controller can be mounted in two position:

- bottom mounting (as per the complete system);
- side mounting.

To install the controller proceed as described in the following procedures.

Bottom Mounting

See the following figure.

- 1** Turn the pump upside-down;
- 2** Screw the 4 fixing studs (provided with the accessories bag) into the holes on the pum bottom;
- 3** Place the controller on the studs aligning the 4 rubber holes on the top of the controller case;
- 4** Press the controller towards the pump;
- 5** Turn the pump up again;
- 6** Plug the line cord and connect the pump cable to the pump;
- 7** Connect the mating connector with the jumper on the interlock signal to start the pump.

17 Technical Information

TwisTorr 74 FS On Board Controller Installation

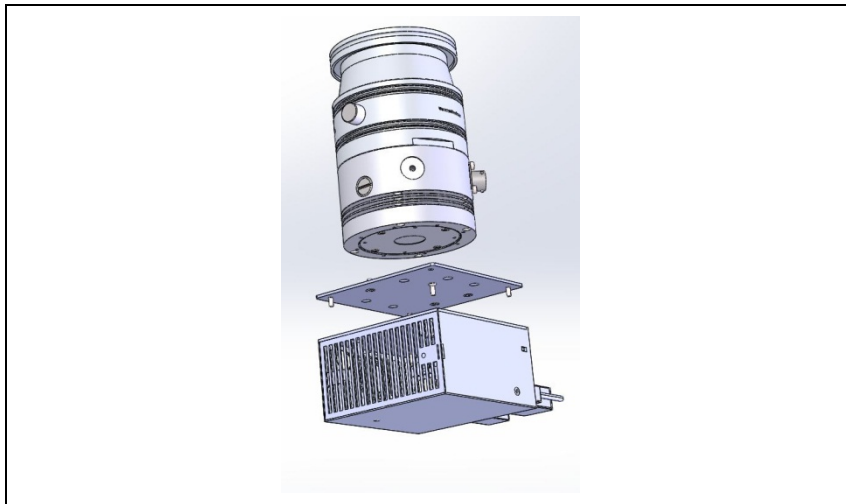


Figure 40 Mounting 1

Side Mounting

NOTE

The L-shaped bracket (P/N X3500-68006) is available as an option.

See the following figure.

- 1** Screw the 4 fixing studs (provided with the accessories bag) in the holes on the L-shaped bracket;
- 2** Place the controller on the studs aligning the 4 rubber holes on the controller top;
- 3** Press the controller towards the bracket;
- 4** Turn the pump upside-down;
- 5** Place the bracket on the pump bottom;
- 6** Screw a little bit the three screws M4 in the holes on the pump bottom;
- 7** Rotate the bracket until the controller is on the chosen position;

- 8** Tighten the screws properly;
- 9** Turn the pump up again;
- 10** Plug the line cord and connect the pump cable to the pump;
- 11** Connect the mating connector with the jumper on the interlock signal to start the pump.

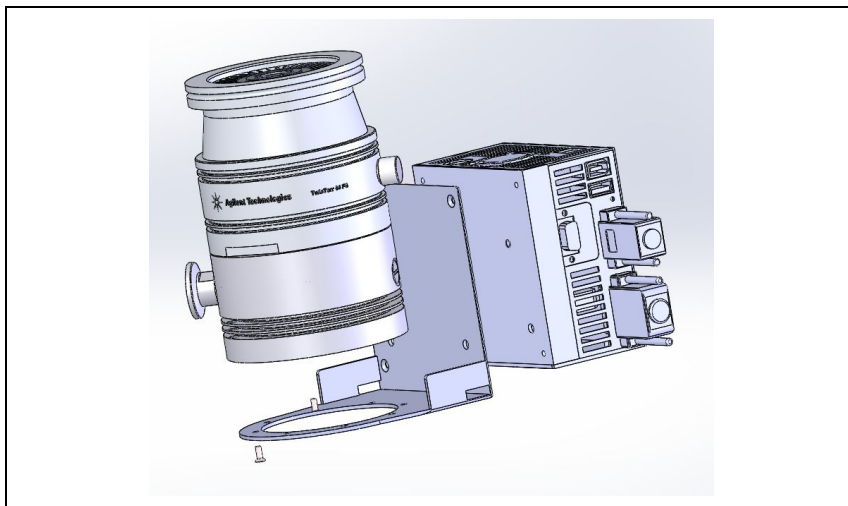


Figure 41 Mounting 2

Connection of the High Vacuum Flange

In the case the rotor is suddenly blocked, the torque arising from the system and the high vacuum flange must be absorbed. Only the components listed in the following can be used to fasten the pump to the high vacuum flange.

WARNING!



It is mandatory to connect the pump in such a way to withstand the torque specified in the "INSTRUCTION FOR USE" section. Specifically pay attention to clamp design, material of clamps and bolts and bolt fixing torque.

WARNING!



If the pump is fastened on a vacuum chamber with different flange Agilent there exists the chance of twisting or tearing-off in case the rotor is suddenly blocked. The combination of different flange types is not allowed in any case. Agilent Vacuum will not accept any liability for all damages resulting from not allowed fastening.

Installation of ISO-K Flange

For ISO-K flange connections, fix the two flanges with the clamps as shown in the following figure (a protective screen can optionally be used).

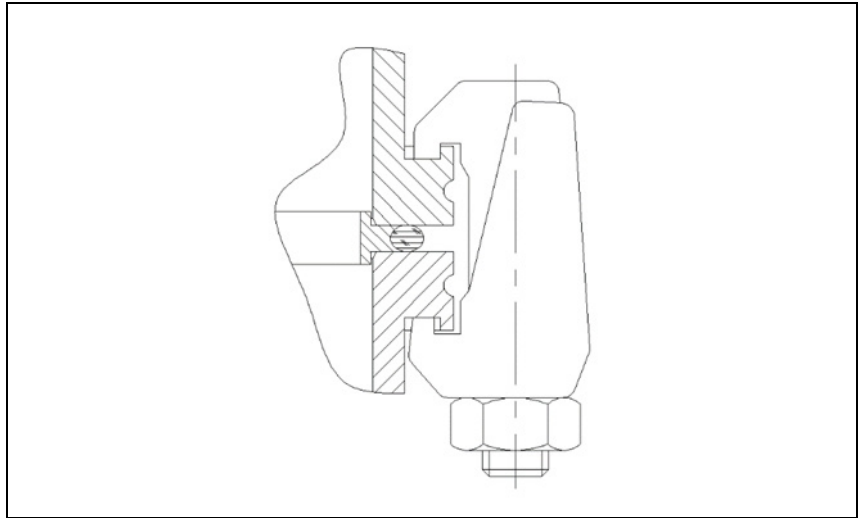


Figure 42

Use the required number of claw clamps:

- 4 for ISO 63 flange

Tighten the claw clamps with a torque of 22 Nm.

Installation of ConFlat Flange (CFF 2 ¾)

For ConFlat flange connections, fix the two flanges with the screws as shown in the following figures (a protective screen can optionally be used).

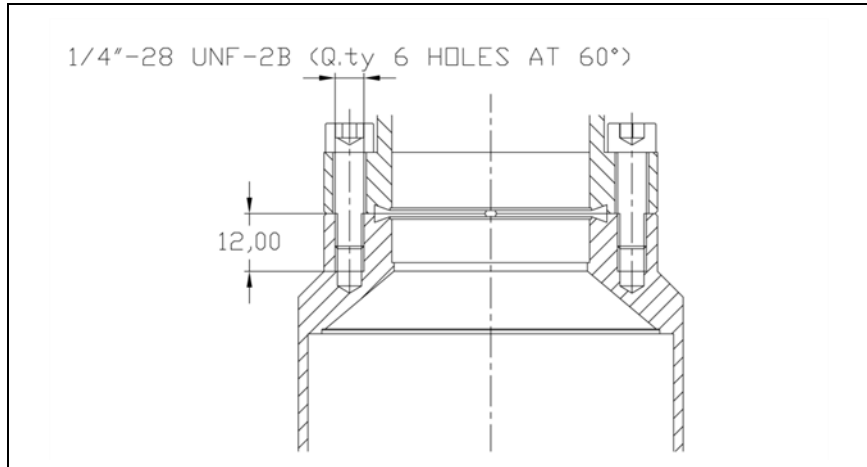


Figure 43

Installation of ConFlat Flange (CFF 4 ½)

For ConFlat flange connections, fix the two flanges with the screws as shown in the following figures (a protective screen can optionally be used).

For ConFlat flange connections, in case of a flange with np-treaded holes fix the two flanges with the screws using the mounting kit X3502-68007, as shown in the following figures (a protective screen can optionally be used).

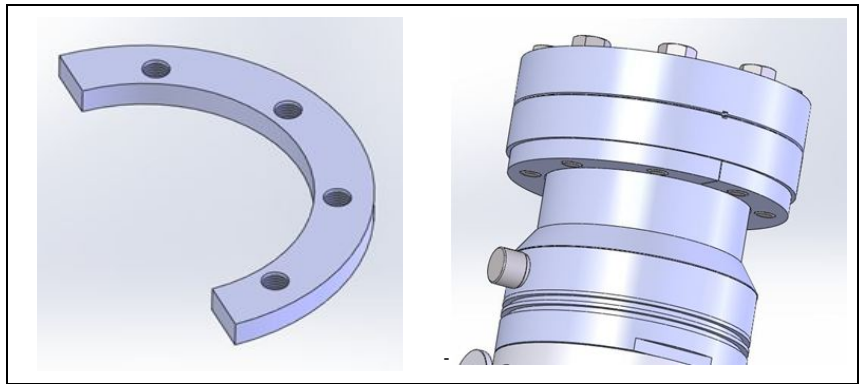


Figure 44 Installation for CFF 4,5"

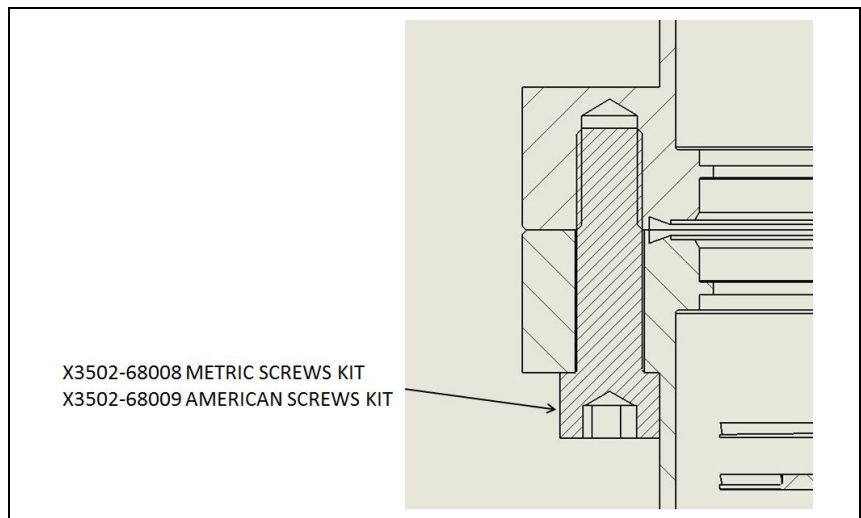


Figure 45 Installation for CFF 4,5"

Use the required number of screws:

- 6 for CFF 2.75"
- 8 for CFF 4.5"

17 Technical Information

Connection of the High Vacuum Flange

Attach the units and tighten each screw in turn. Repeat the sequential tightening until the flange faces meet. Tighten the screws with a torque of 20 Nm.

For ConFlat flange connections we recommend using Agilent hardware.

To facilitate assembly and dismantling, apply Felpro C-100 high temperature lubricant to the screw threads protruding from the flange and between the nuts and flange.

Note that, in some cases, the connections can be made only with the bolt in the lower side.

CAUTION!

Exercise care when tightening nuts and bolts to avoid creating dents in the envelope as this may cause the pump rotor to lock.

How to install the turbo pump without the mounting kit X3502-68007:

- Contra flange has threaded holes, fixing screws must be inserted from the bottom flange (screws longer than 30 mm can not be used). It is possible to order special wrench screws: metric X3502-68008 or American screws X3502-68009.

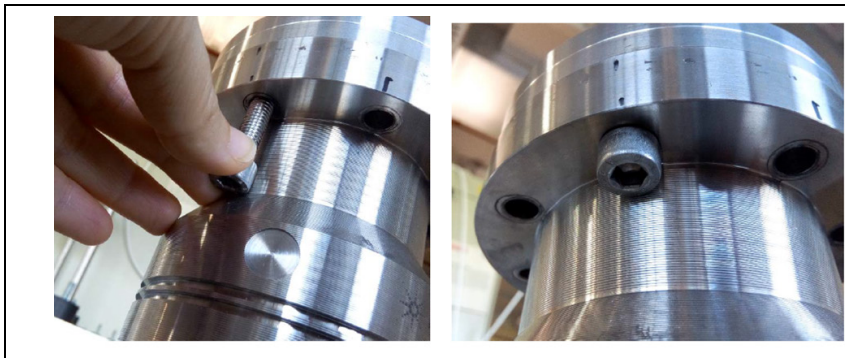


Figure 46 | Installation for CFF 4,5"

- Both flanges have through holes and nuts are tightened by the envelope flange. The nut can not rotate freely and it is mandatory to use a key to hold it while tightening the fixing screws.

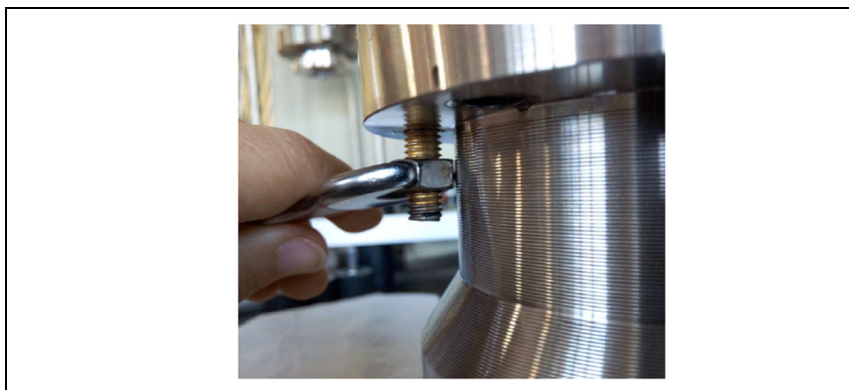


Figure 47 II Installation for CFF 4,5"

CAUTION!

The nut can cause damages not only from an aesthetical point of view but it may as well have consequences on pump operation.

- Contra flange has through holes, screws cannot be inserted from the bottom as there is not enough space to accommodate them.



Figure 48 II Installation for CFF 4,5"

17 Technical Information

Connection of the High Vacuum Flange

CAUTION!

The screws must be inserted from the top flange while nuts must be fixed below the flange.

Connection of the Fore-Vacuum Pump

A flange KF 16 NW is available to connect the TwisTorr 74 FS pump to the forevacuum pump. A hose or vacuum approved pipe can be used. If a rigid pipe is used, any vibration generated by the mechanical pump must be eliminated through the use of bellows.

NOTE

The TwisTorr 74 FS pump is characterized by its high compression ratio also for oil vapors. When using a mechanical oil sealed pump, it is advisable to install a suitable trap between the turbopump and the fore vacuum pump in order to prevent oil back streaming.

Pump Used with Corrosive Gases

To prevent damage to the bearings, an inert gas must flow into the pump body around the upper bearing towards the forevacuum line. To supply the inert purge gas (e. g. nitrogen) to the pump through the purge port, connect a gas purge valve between the pressure regulator and the pump.

Adjust the pressure regulator in order to read a gas flow rate of 0.1 to 0.8 mbar l/s.

CAUTION!

To prevent bearing damage, Agilent suggests a minimum purge gas flow rate of 10 sccm (0.17 mbar l/s). This value can be exceeded, according to the process requirements. Please contact Agilent for specific applications.

The purge gas throughput with the recommended forepump of 15 m³/h (11 CFM) allows achieving a high vacuum pressure in the 10-8 mbar range. The recommended gas flow maintains a pressure into the pump body higher than the forevacuum pressure. The recommended procedure to vent the system and the pump avoiding the contact between the pump bearings and the corrosive gas is described in the following points:

- 1 Close the corrosive gas flow into the system.
- 2 Leave the Turbo pump and the backing pump running and the purge gas flowing, wait for enough time to evacuate the corrosive gas from the system.
- 3 Turn off the Turbopump.
- 4 Open the Turbo vent port slowly until to reach atmospheric pressure in the system.
- 5 When the Turbo pump and the backing pump are stopped and the system is at atmospheric pressure, for a better bearing protection it is advisable to leave the purge gas flowing into the Turbo pump, with the chamber or the Turbo vent valve opened, to avoid system overpressure. If the vent valve can't be kept opened, the backing pump should be left operating.

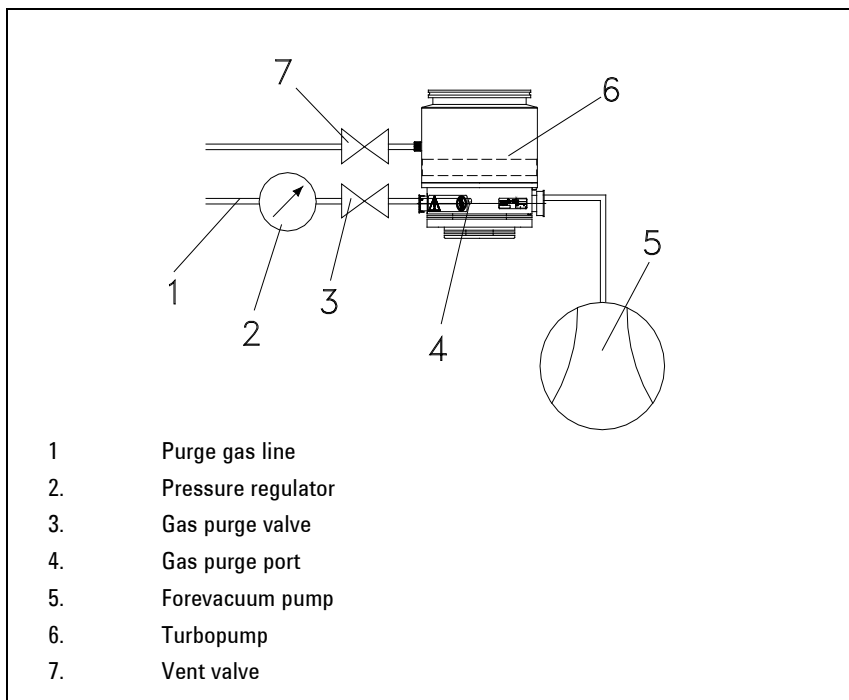


Figure 49 Purge layout

17 Technical Information

Venting Procedure (without Contaminants and/or Corrosive Gases) – Manual Venting

Venting Procedure (without Contaminants and/or Corrosive Gases) – Manual Venting

- 1 Close the process gases flowing into the system before sending the STOP signal to the TwisTorr 74 FS pump.
- 2 Close the gate valve connecting the Turbo pump to the chamber (if present).
- 3 Switch off the backing pump or close a fore-vacuum valve (if present).
- 4 Turn off the Turbo pump by sending the STOP signal to the controller.
- 5 Open the Vent Port (positioned on the envelope) slowly until to reach atmospheric pressure in the system.
- 6 For water cooling: shut off the water supply.

CAUTION!

To safely slow down the pump use a venting flow able to decelerate the rotor at a rate lower than 13-15 Hz/s. Higher deceleration rates could damage the bearings.

Pump Used in Presence of Magnetic Fields

Magnetic fields induce eddy currents in the rotor of a turbomolecular pump that tend to oppose to its rotation.

The result is increased electrical power consumption by the motor, most of which is dissipated in the rotor.

Since the rotor is not in contact with the stator the above power can leave the rotor mainly by radiation and hence the rotor may be overheated while static parts of the pump remain cool.

This effect is strongly dependant from the intensity, time function and distribution of the magnetic field.

In general, therefore, an increase in pump current can be expected.

If this increase is lower than 50 % of the current value drawn by the motor in high vacuum operation, no particular problem should be expected.

However if the effect is greater, then the application should be carefully reviewed by Agilent's specialist. As a matter of fact, in case of high magnetic fields, also important forces might be generated and applied to the rotor.

Accessories and Spare Parts

Tab. 4 Accessories and spare parts

Description	Part Numbers
Mains cable NEMA plug, 3 m long	9699958
Mains cable European plug, 3 m long	9699957
Serial cable and A-plus software	9699883
Inlet screen ISO63	X3502-68001
Inlet screen CFF4.5"	X3502-68000
Inlet screen, KF40	9699309
Inlet screen CFF 2.75"	9699328
Metal Water Cooling kit	X3502-68002
Plastic Water cooling kit	X3502-68003
Air cooling kit (0,5 m cable)	9699290
Air cooling kit extension cable (5 m)	9699940
Controller side mounting bracket	X3502-68006
Vibration isolator ISO 63	9699375
Vibration isolator CFF 4.5"	9699376
Vent Valve N.O. 0,5 mm orifice (0,5 m cable)	9699844
Vent Valve extension cable (5 m)	9699941
Vent Screw M5	X3502-68005
Purge Screw	X3502-68004
Purge valve 10 SCCM NW16KF - M12	9699239
Purge valve 10 SCCM ¼ Swagelok - M12	9699240
Purge valve 20 SCCM NW16KF - M12	9699241
Purge valve 20 SCCM ¼ Swagelok - M12	9699242
Purge valve 10 SCCM ¼ Swagelok - ¼ Swagelok	9699232
Purge valve 20 SCCM ¼ Swagelok - ¼ Swagelok	9699236
CFF4.5 Mounting kit	X3502-68007
Metric screws kit	X3502-68008
American screws kit	X3502-68009
Pump extension cable (3 m)	9699942



Vacuum Products Division

Dear Customer,

Thank you for purchasing an Agilent vacuum product. At Agilent Vacuum Products Division we make every effort to ensure that you will be satisfied with the product and/or service you have purchased.

As part of our Continuous Improvement effort, we ask that you report to us any problem you may have had with the purchase or operation of our products. On the back side you find a Corrective Action request form that you may fill out in the first part and return to us.

This form is intended to supplement normal lines of communications and to resolve problems that existing systems are not addressing in an adequate or timely manner.

Upon receipt of your Corrective Action Request we will determine the Root Cause of the problem and take the necessary actions to eliminate it. You will be contacted by one of our employees who will review the problem with you and update you, with the second part of the same form, on our actions.

Your business is very important to us. Please, take the time and let us know how we can improve.

Sincerely,

Giampaolo LEVI

*Vice President and General Manager
Agilent Vacuum Products Division*

Note: Fax or mail the Customer Request for Action (see backside page) to Agilent Vacuum Products Division (Torino) – Quality Assurance or to your nearest Agilent representative for onward transmission to the same address.

CUSTOMER REQUEST FOR CORRECTIVE / PREVENTIVE / IMPROVEMENT ACTION

TO: AGILENT VACUUM PRODUCTS DIVISION TORINO – QUALITY ASSURANCE FAX

N°: XXXX-011-9979350

ADDRESS: AGILENT TECHNOLOGIES ITALIA S.p.A. – Vacuum Products Division –

Via F.lli Varian, 54 – 10040 Leini (TO) – Italy

E-MAIL: vpd-qualityassurance_pdl-ext@agilent.com

NAME	COMPANY	FUNCTION
ADDRESS:		
TEL. N° : FAX N° :		
E-MAIL:		
PROBLEM / SUGGESTION :		
REFERENCE INFORMATION (model n°, serial n°, ordering information, time to failure after installation, etc.):		
DATE		
CORRECTIVE ACTION PLAN / ACTUATION (by AGILENT VPD)		LOG N°

XXX = Code for dialing Italy from your country (es. 01139 from USA; 00139 from Japan, etc.)



**Vacuum Products Division
Instructions for returning products**

Dear Customer,

Please follow these instructions whenever one of our products needs to be returned.

Complete the attached **Request for Return form** and send it to Agilent Technologies (see below), taking particular care to include the completed **Health and Safety** declaration Section. No work can be started on your unit until we receive a completed copy of this form.

After evaluating the information, Agilent Technologies will provide you with a **Return Authorization (RA) number** via email or fax, as requested. Note: Depending on the type of return, a Purchase Order may be required at the time the **Request for Return** is submitted. We will quote any necessary services (evaluation, repair, special cleaning, eg).

Product preparation

- Remove all accessories from the core product (e.g. inlet screens, vent valves).
- Prior to shipment and if applicable for your product, drain any oils or other liquids, purge or flush all gasses, and wipe off any excess residue.
- If ordering an Advance Exchange product, please use the packaging from the Advance Exchange to return the defective product.
- Seal the product in a plastic bag, and package product carefully to avoid damage in transit. You are responsible for loss or damage in transit.
- Include a copy of the Health and Safety Declaration in the shipping documentation on the outside of the shipping box of your returning product.
- Clearly label package with RA number. Using the shipping label provided will ensure the proper address and RA number are on the package. Packages shipped to Agilent without a RA clearly written on the outside cannot be accepted and will be returned.
- Return only products for which the RA was issued.

Shipping

- Ship to the location specified on the printable label, which will be sent, along with the RA number, as soon as we have received all of the required information. Customer is responsible for freight charges on returning product.
- Return shipments must comply with all applicable Shipping Regulations (IATA, DOT, ADR, etc.) and carrier requirements.

RETURN THE COMPLETED REQUEST FOR RETURN FORM TO YOUR NEAREST LOCATION:

EUROPE:

Fax: 00 39 011 9979 330

Fax Free: 00 800 345 345 00

Toll Free: 00 800 234 234 00

vpt-customer@agilent.com

NORTH AMERICA:

Fax: 1 781 860 9252

Toll Free: 800 882 7426

vpl-ra@agilent.com

PACIFIC RIM:

please visit our website for individual
office information

<http://www.agilent.com>



TERMS AND CONDITIONS

Please read the terms and conditions below as they apply to all returns and are in addition to the Agilent Technologies Vacuum Product Division – Products and Services Terms of Sale.

- Unless otherwise pre-negotiated, customer is responsible for the freight charges for the returning product. Return shipments must comply with all applicable **Shipping Regulations** (IATA, DOT, etc.) and carrier requirements.
- Agilent Technologies is not responsible for returning customer provided packaging or containers.
- Customers receiving an Advance Exchange product agree to return the defective, rebuildable part to Agilent Technologies **within 15 business days**. Failure to do so, or returning a non-rebuildable part (crashed), will result in an invoice for the non-returned/non-rebuildable part.
- Returns for credit toward the purchase of new or refurbished Products are subject to prior Agilent approval and may incur a restocking fee. Please reference the original purchase order number.
- Units returned for evaluation will be evaluated, and a quote for repair will be issued. If you choose to have the unit repaired, the cost of the evaluation will be deducted from the final repair pricing. A Purchase Order for the final repair price should be issued within 3 weeks of quotation date. Units without a Purchase Order for repair will be returned to the customer, and the evaluation fee will be invoiced.
- Products returned that have not been drained from oil will be disposed.
- A Special Cleaning fee will apply to all exposed products
- If requesting a calibration service, units must be functionally capable of being calibrated.

Vacuum Products Division Request for Return Form

Customer information		
Company :	Contact Name:	
Address:	Tel:	Fax:
	Email:	

Equipment			
Product description	Agilent PartNo	Agilent Serial No	Original Purchasing Reference
Failure description		Type of process (for which the equipment was used)	

Type of return	
☐ Non Billable ☐ Billable ➔ New PO # (hard copy must be submitted with this form): _____	
☐ Exchange ☐ Repair ☐ Upgrade ☐ Consignment/Demo ☐ Calibration ☐ Evaluation ☐ Return for Credit	

Health and safety		Substances (please refer to MSDS forms)			
The product has been exposed to the following substances: (by selecting 'YES' you MUST complete the table to the right)		* Agilent will not accept delivery of any product that is exposed to radioactive, biological, explosive substances or dioxins, PCB's without written evidence of decontamination.			
		Trade name	Chemical name	Chemical Symbol	CAS Number
Toxic	☐ YES ☐ NO				
Harmful	☐ YES ☐ NO				
Corrosive	☐ YES ☐ NO				
Reactive	☐ YES ☐ NO				
Flammable	☐ YES ☐ NO				
Explosive (*)	☐ YES ☐ NO				
Radioactive (*)	☐ YES ☐ NO				
Biological (*)	☐ YES ☐ NO				
Oxidizing	☐ YES ☐ NO				
Sensitizer	☐ YES ☐ NO				
Other dangerous substances	☐ YES ☐ NO				

Goods preparation	
If you have replied YES to one of the above questions. Has the product been purged? ☐ YES ☐ NO	
If yes, which cleaning agent/method: _____	
Has the product been drained from oil? ☐ YES ☐ NOT APPLICABLE	
I confirm to place this declaration on the outside of the shipping box. ☐	

I declare that the above information is true and complete to the best of my knowledge and belief. I understand and agree to the terms and conditions on page 2 of this document.	
Name:	Authorized Signature:
Position:	
Date:	
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